

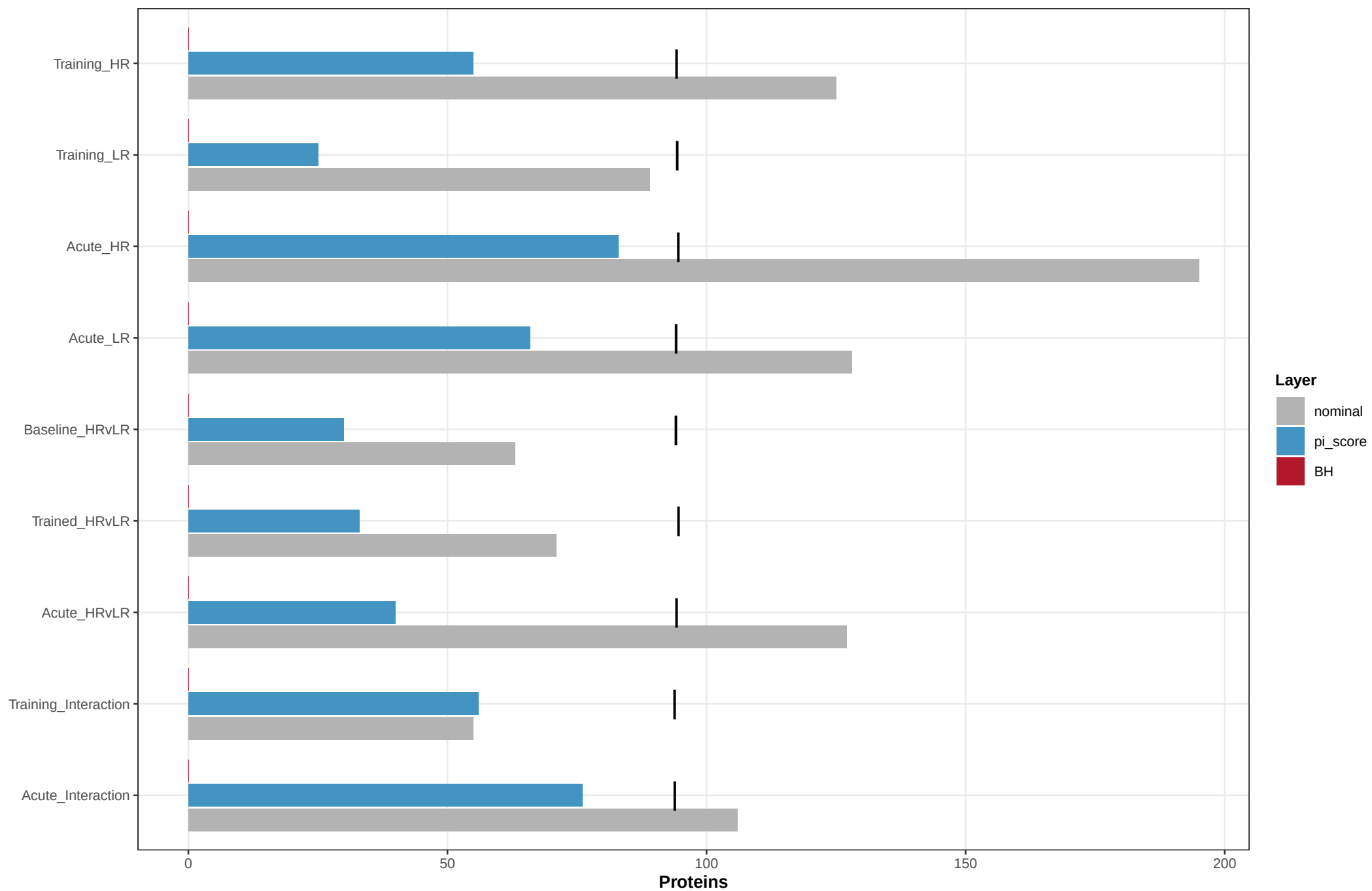
HRvLR 02 Proteins

Generated 2026-09-23

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Proteins called per contrast

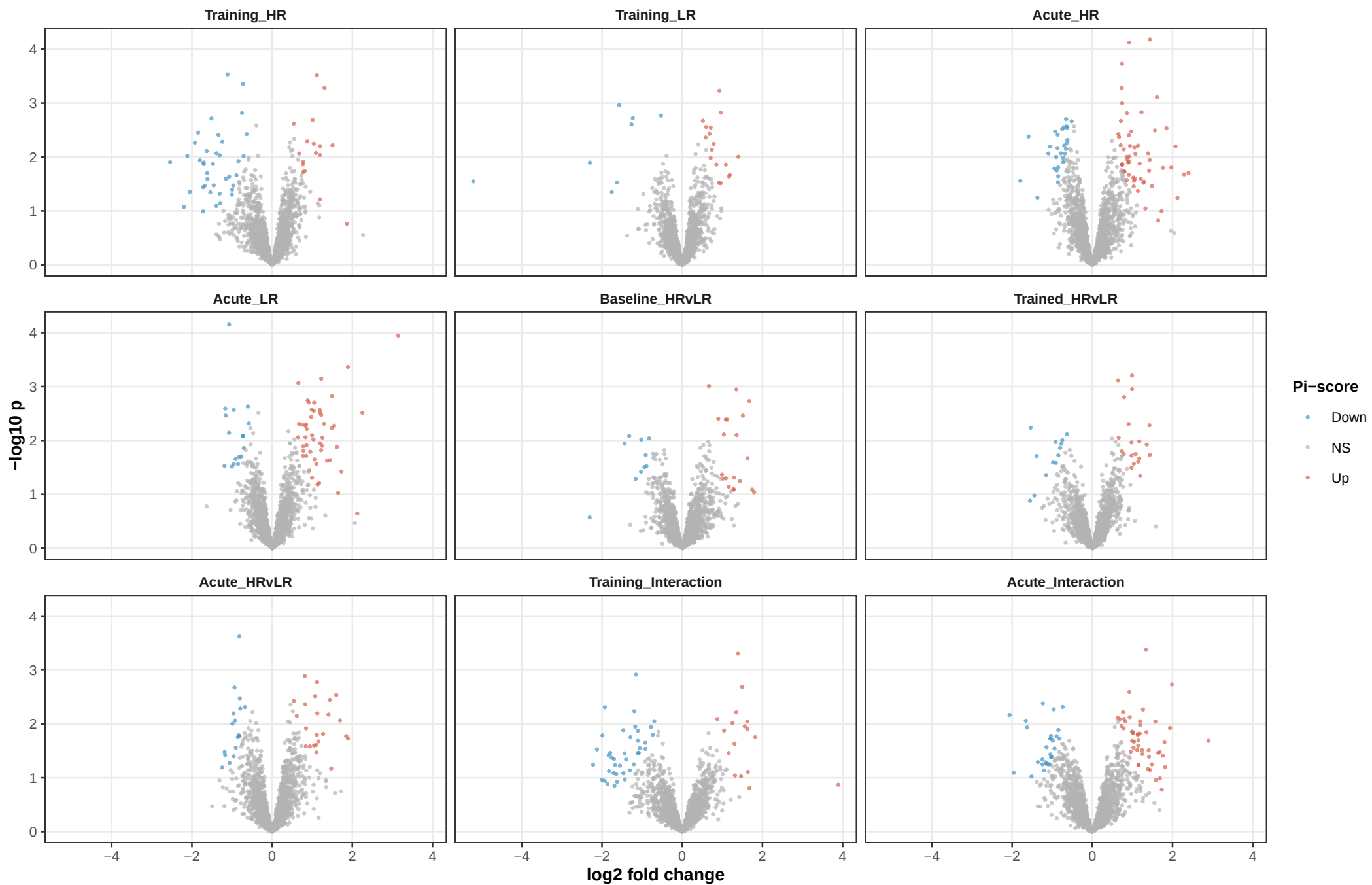
limma via proteoDA, ~ 0 + group + (1 | subject); 1900 proteins; BH within contrast



Bars: proteins at nominal $p < 0.05$, at pi-score < 0.05 ($p^*|\log_2FC|$, Xiao 2014; no FDR control) and at BH < 0.05 . Vertical tick: the count expected at $p < 0.05$ if no protein responded ($0.05 \times \text{proteins tested}$). Data: 02_Proteins/01_Differential/c_data/01_dep_results.csv.

Volcano per contrast

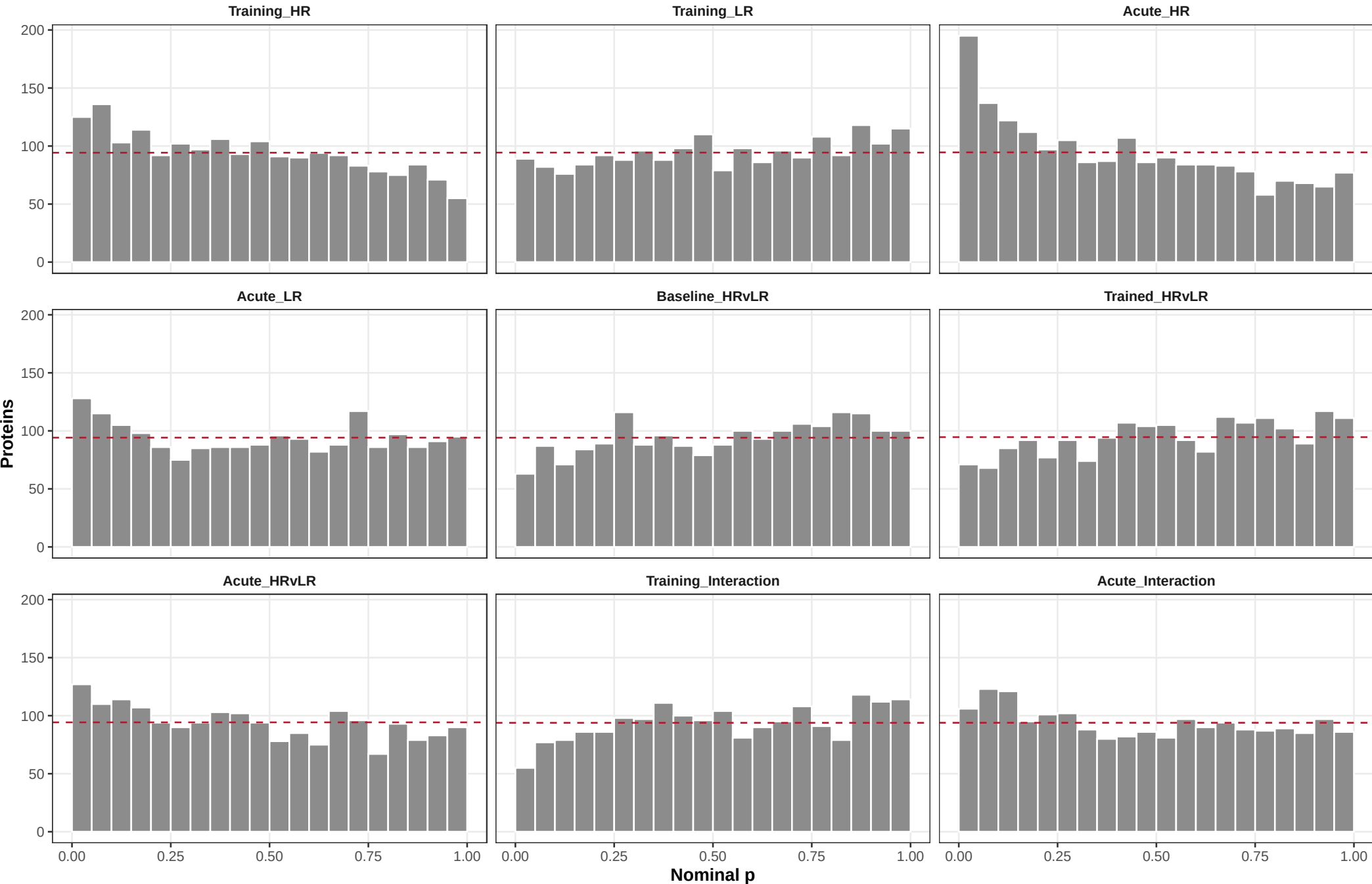
log2 fold change against nominal p, coloured by pi-score call



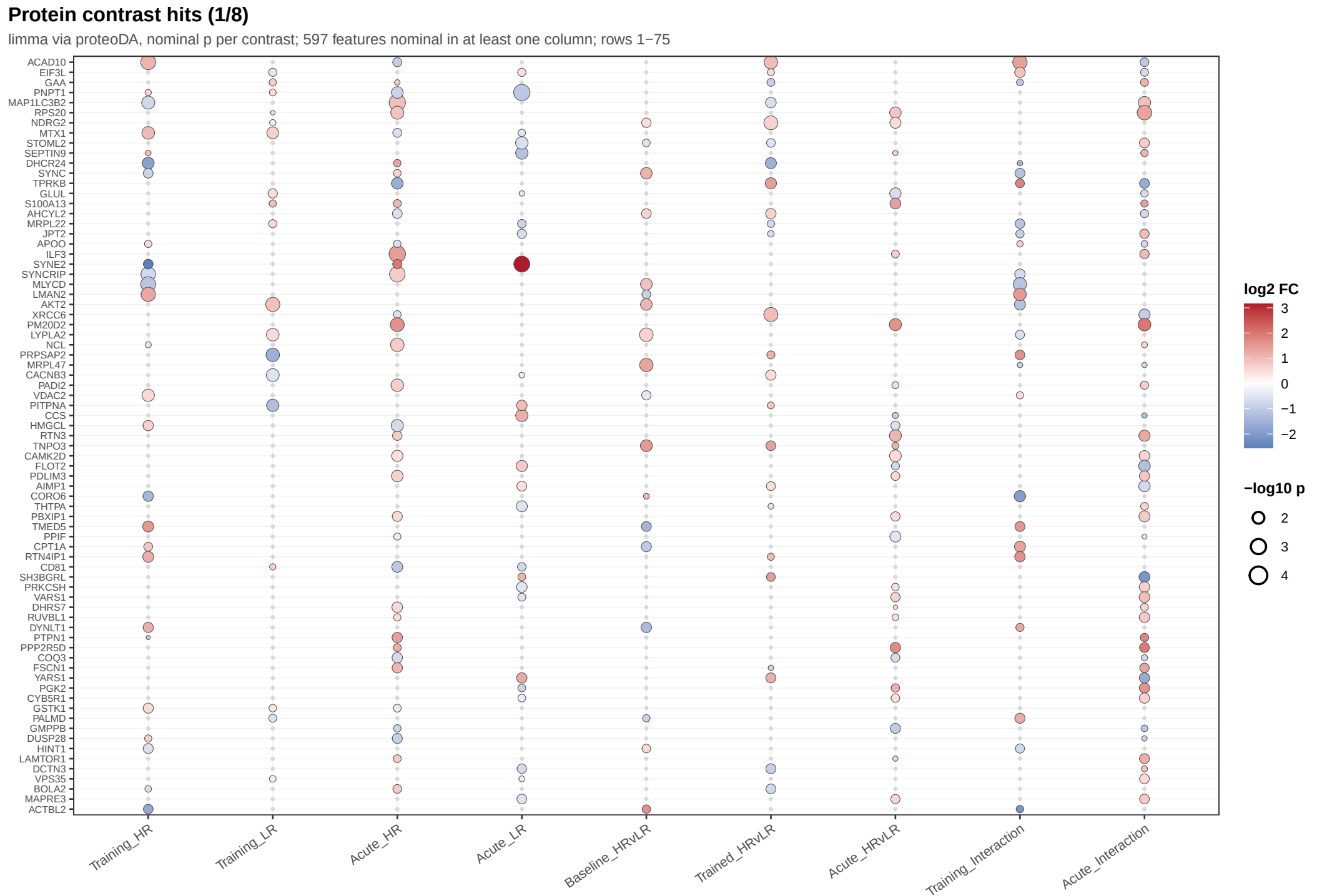
Each point is one protein in one contrast. Colour: up or down at pi-score < 0.05, grey otherwise. Proteins at BH < 0.05 across all contrasts: 0. Data: 02_Proteins/01_Differential/c_data/01_dep_results.csv.

P-value distribution per contrast

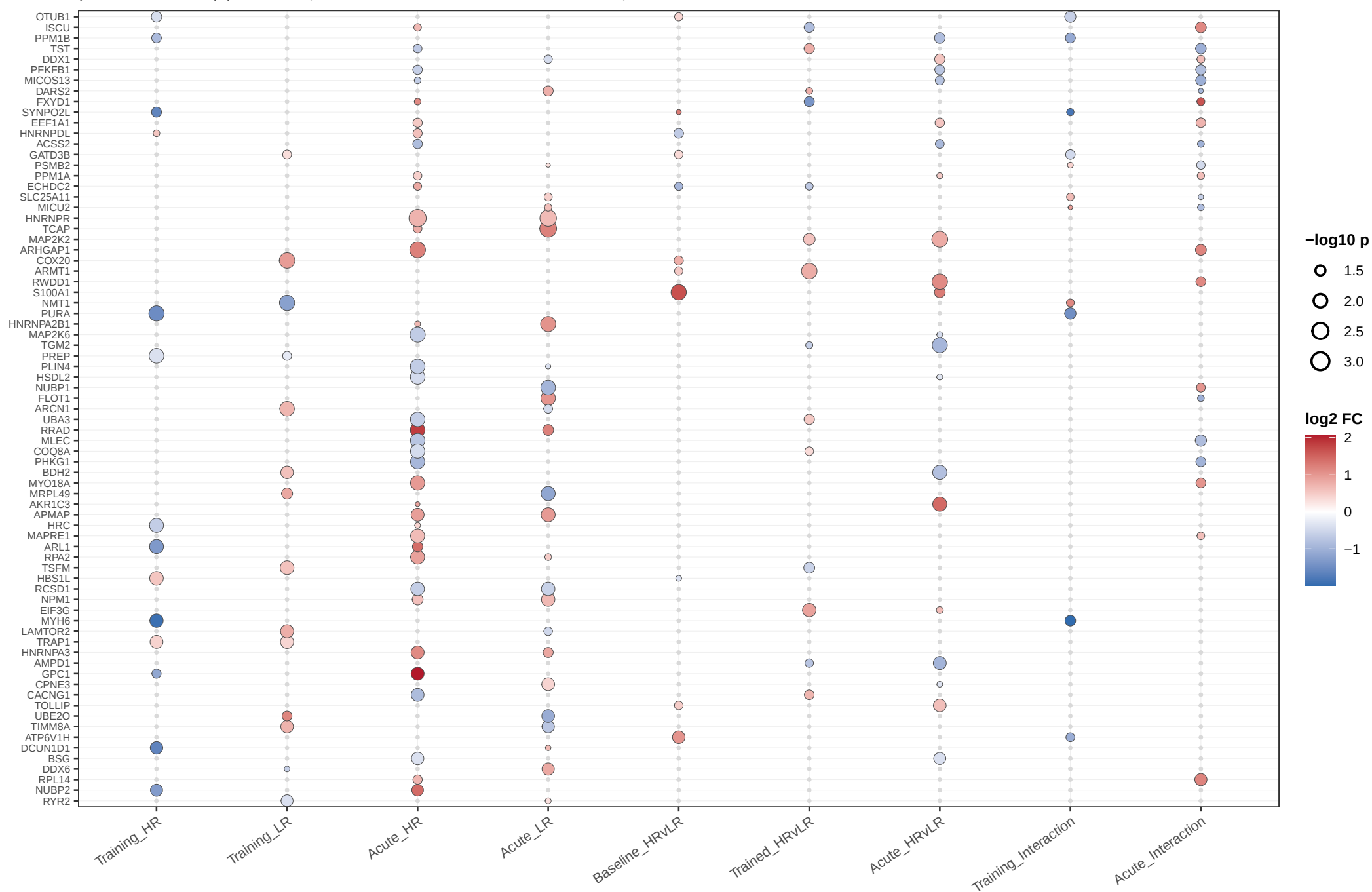
20 bins of width 0.05



A uniform histogram is what a contrast with no signal produces; a spike near zero is signal, a hump near one is over-dispersion or a misspecified variance. Dashed line: the uniform height (tested / 20). Data: 02_Proteins/01_Differential/c_data/01_dep_results.csv.

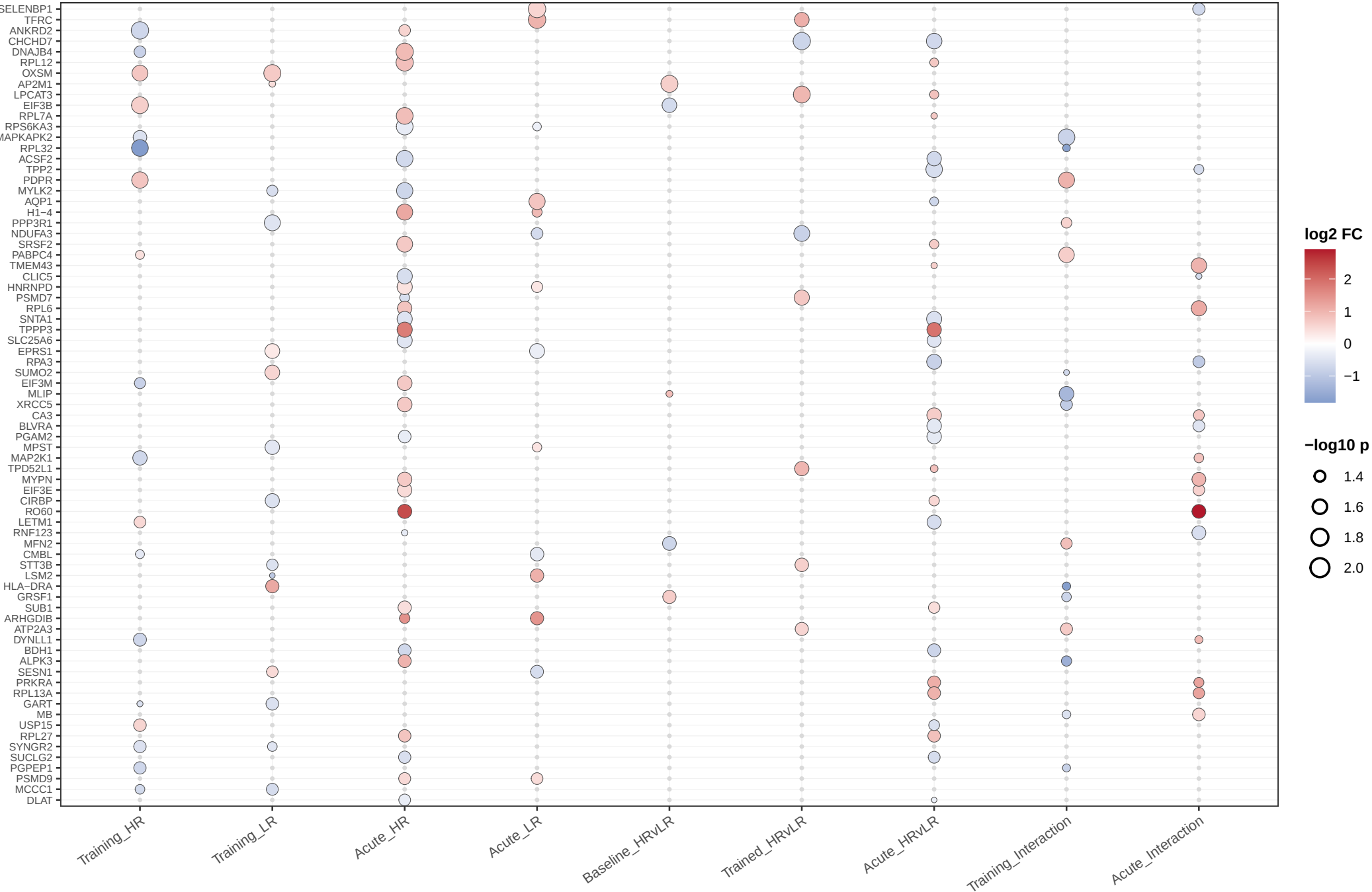


Protein contrast hits (2/8)
 limma via proteoDA, nominal p per contrast; 597 features nominal in at least one column; rows 76–150



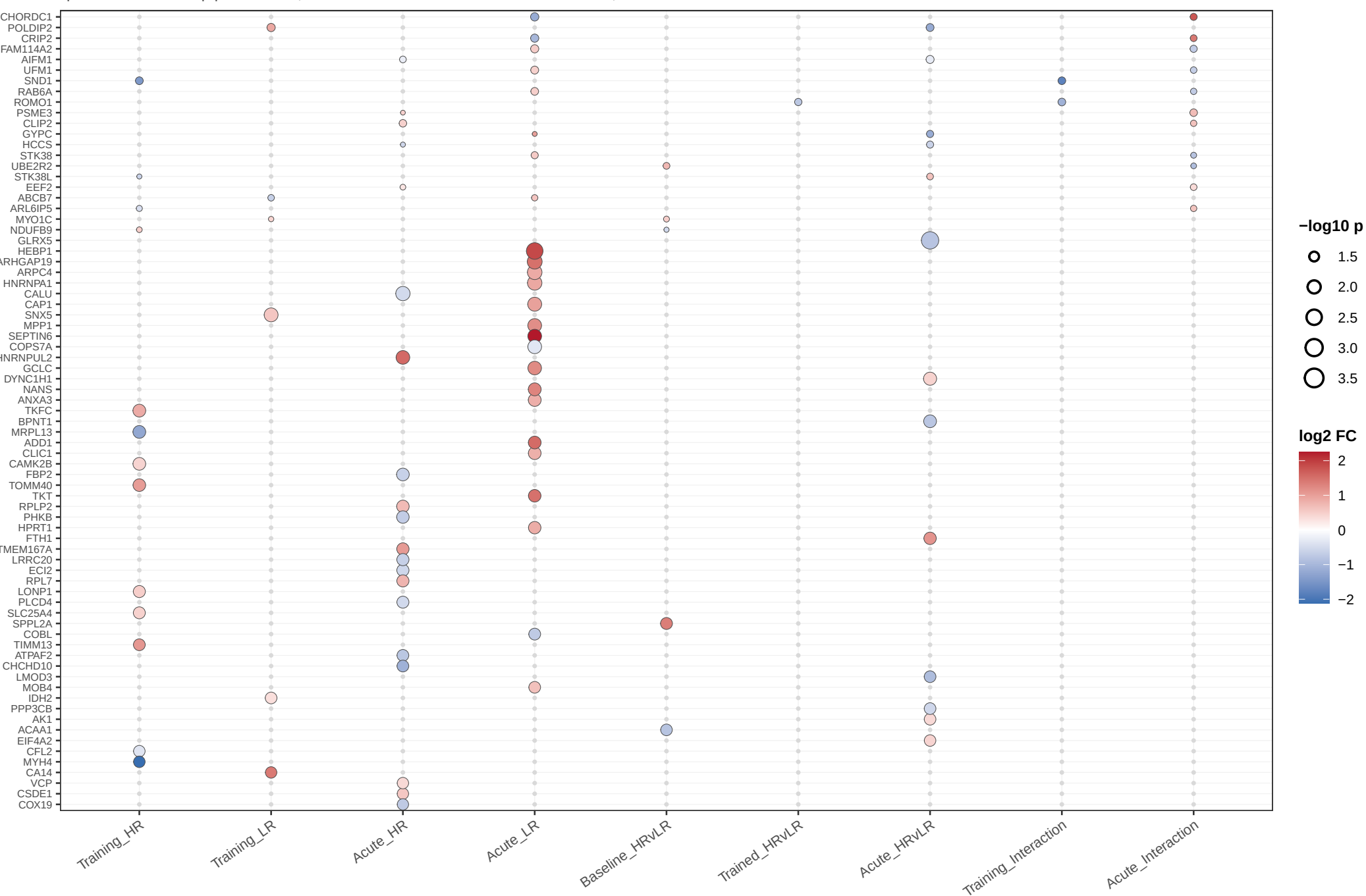
Protein contrast hits (3/8)

limma via proteoDA, nominal p per contrast; 597 features nominal in at least one column; rows 151–225



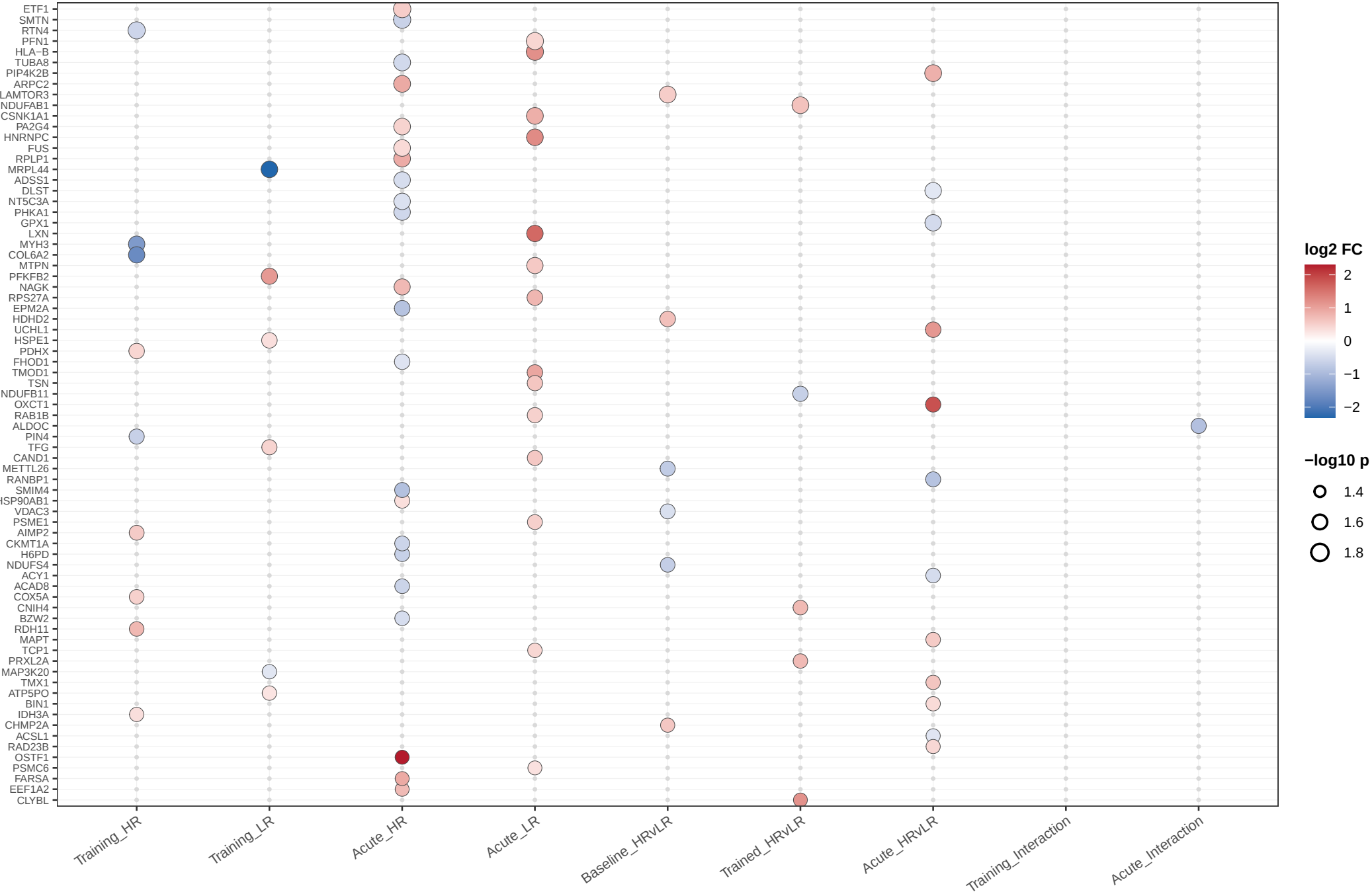
Protein contrast hits (4/8)

limma via proteoDA, nominal p per contrast; 597 features nominal in at least one column; rows 226–300



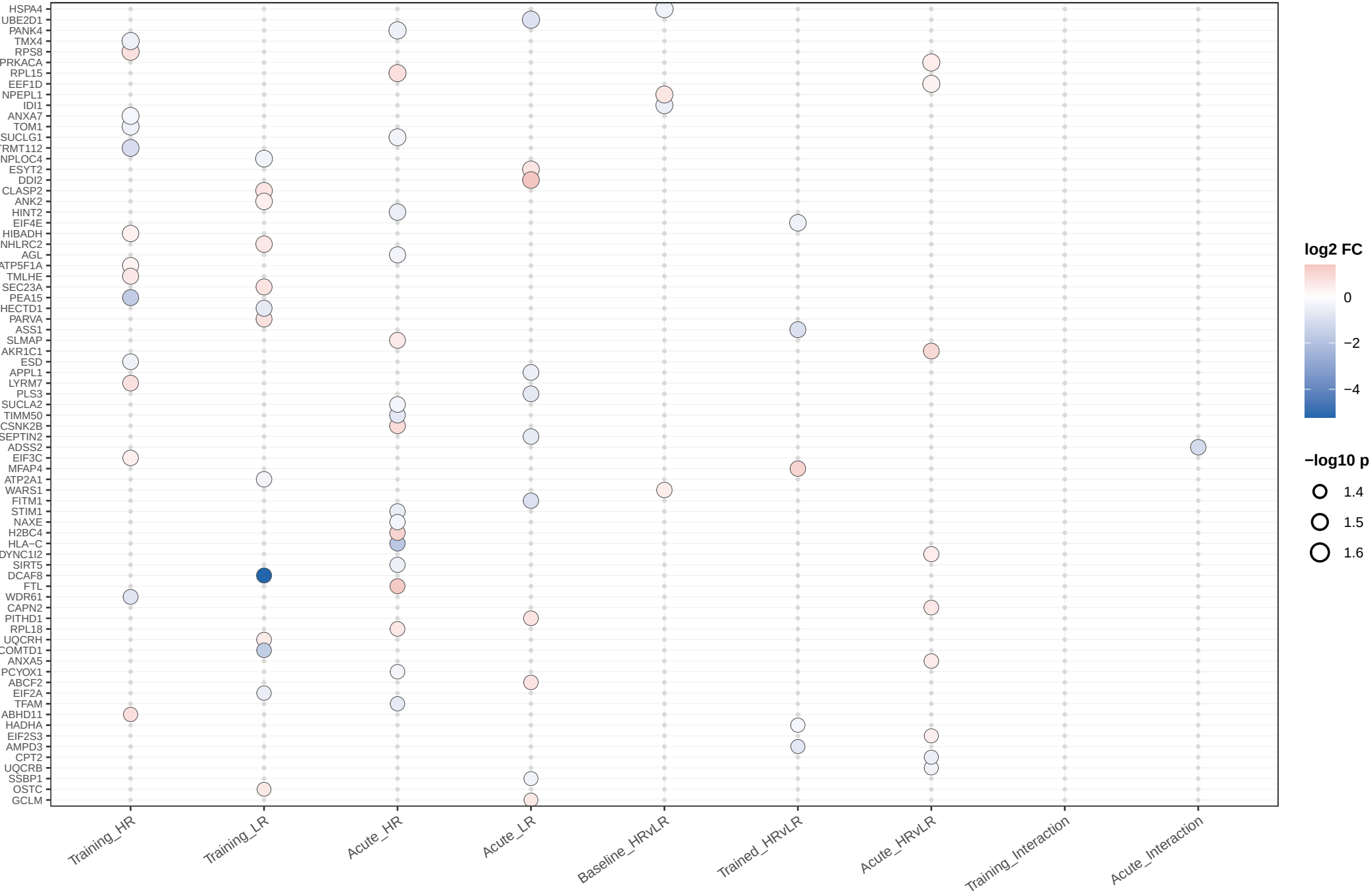
Protein contrast hits (5/8)

limma via proteoDA, nominal p per contrast; 597 features nominal in at least one column; rows 301–375

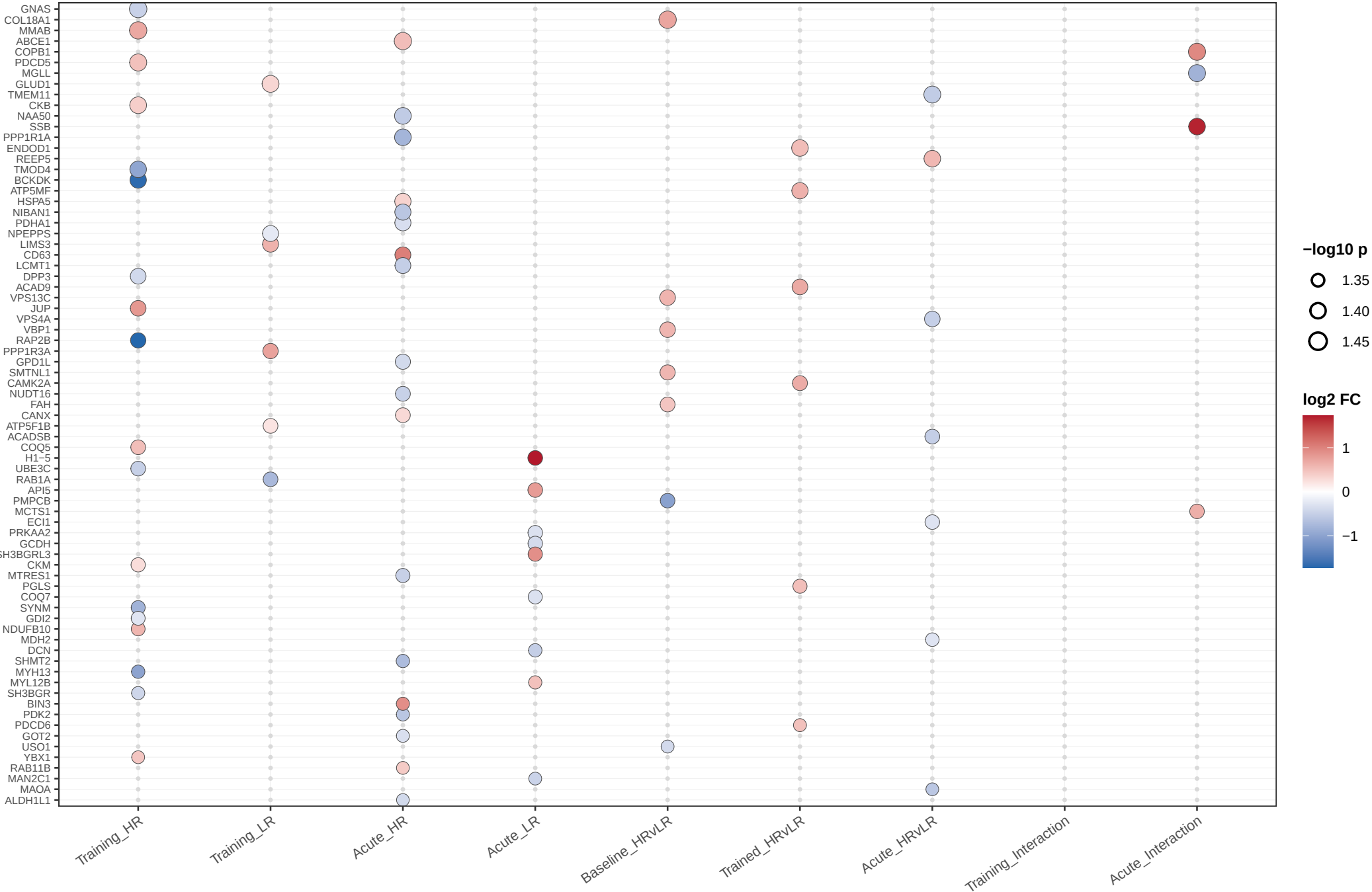


Protein contrast hits (6/8)

limma via proteoDA, nominal p per contrast; 597 features nominal in at least one column; rows 376–450



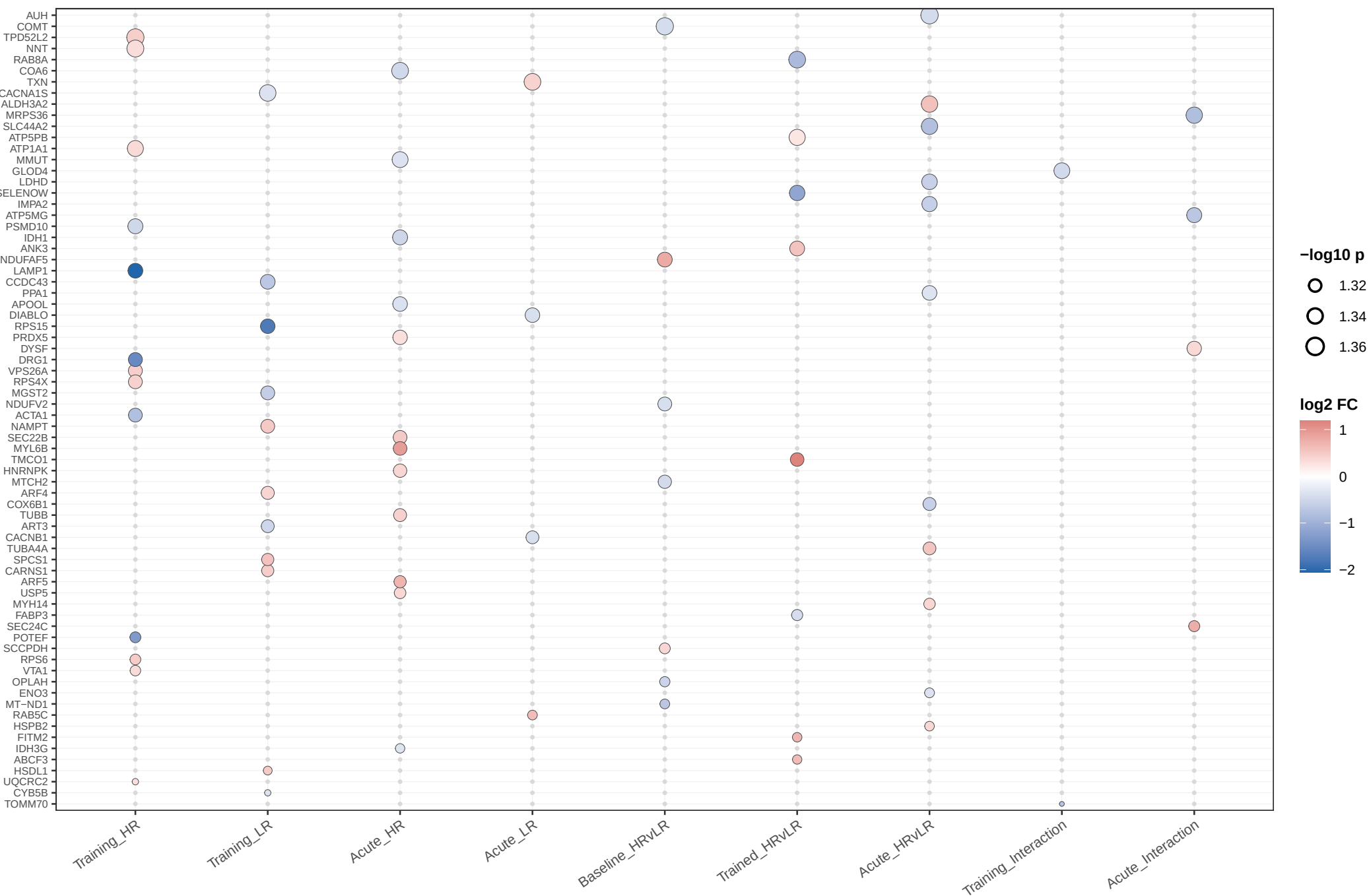
Protein contrast hits (7/8)
limma via proteoDA, nominal p per contrast; 597 features nominal in at least one column; rows 451–525



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = log2 FC, size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/01_Differential/c_data/01_dep_results.xlsx.

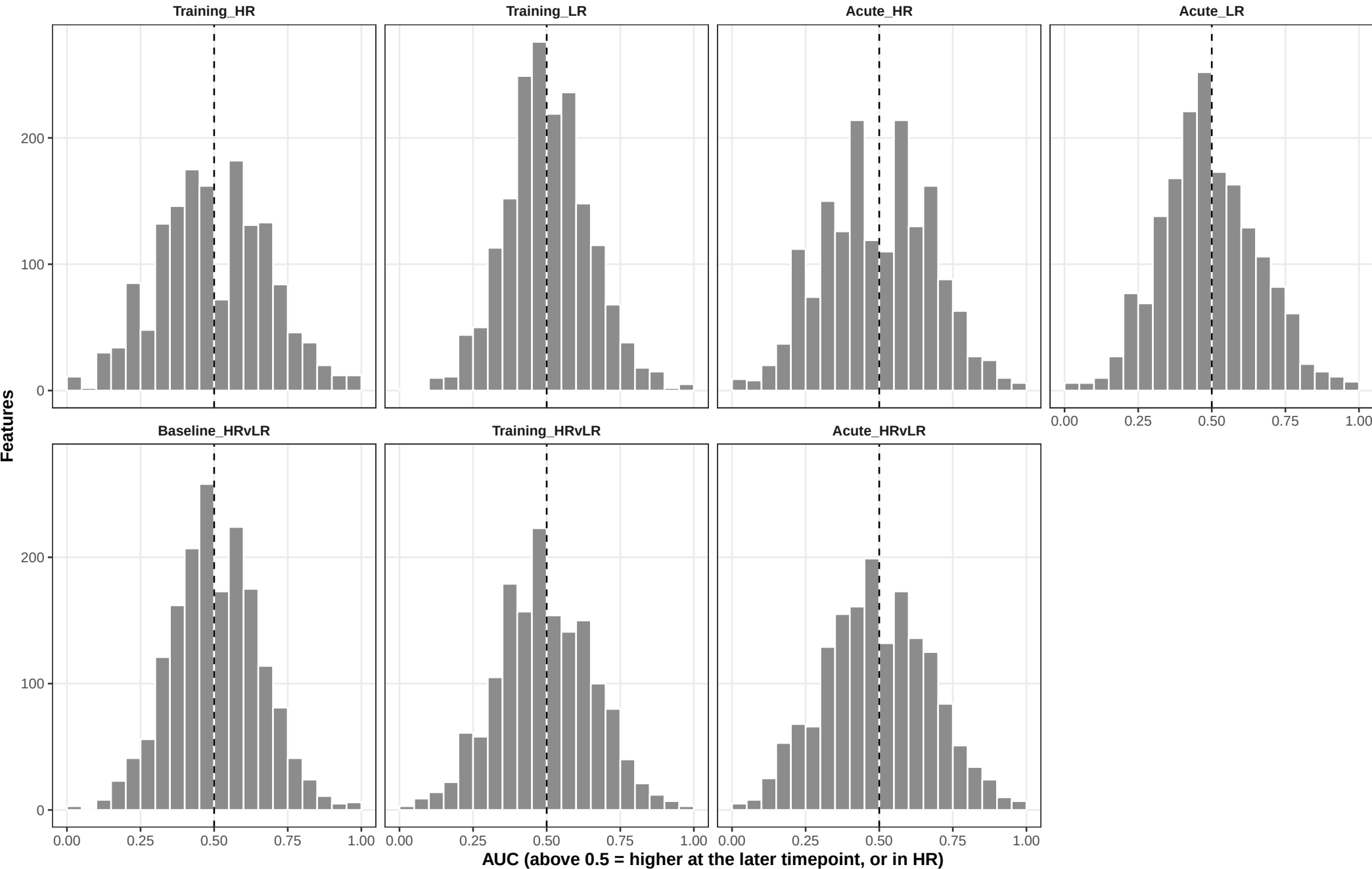
Protein contrast hits (8/8)

limma via proteoDA, nominal p per contrast; 597 features nominal in at least one column; rows 526–597



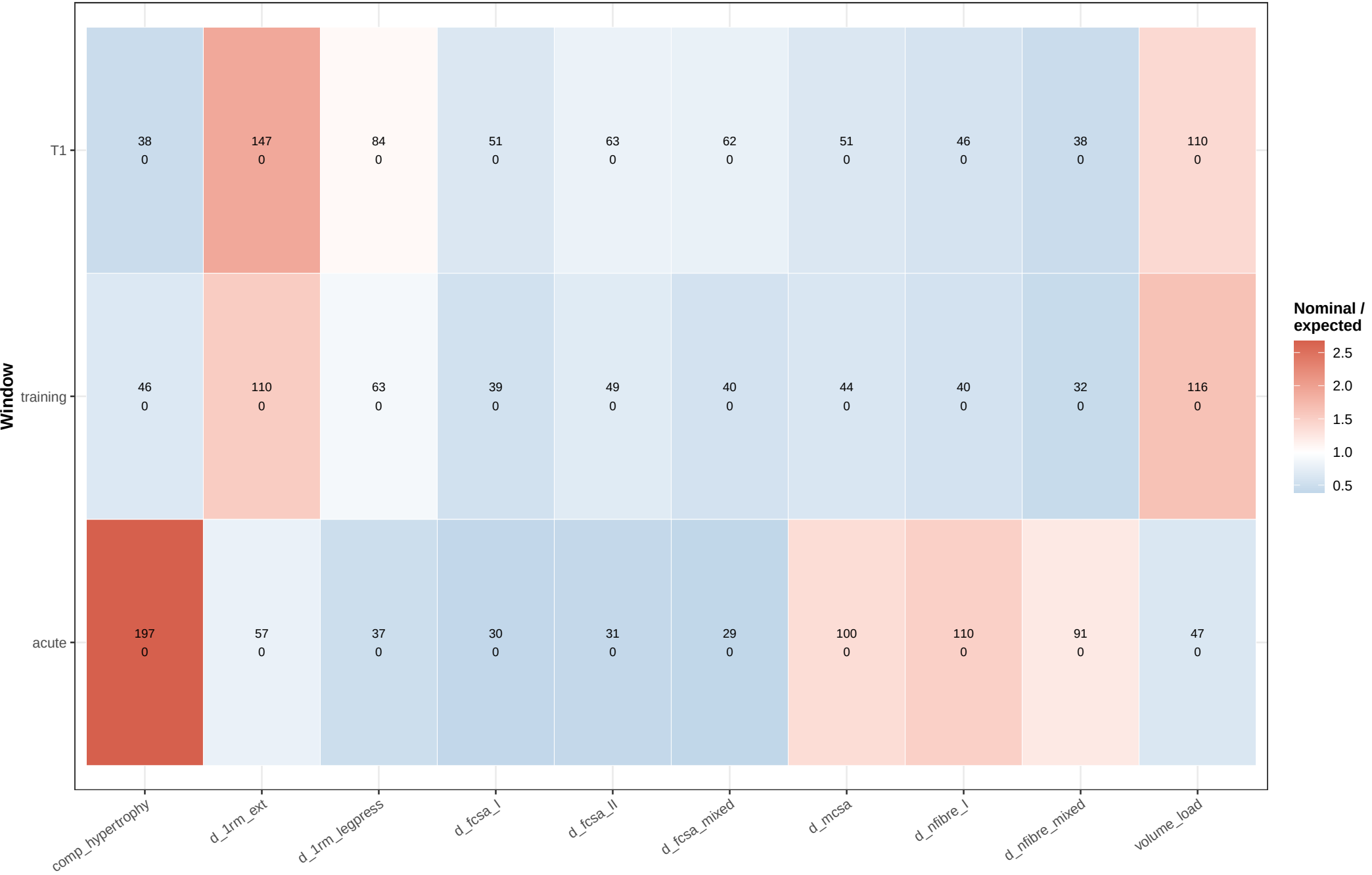
Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = $\log_2 FC$, size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/01_Differential/c_data/01_dep_results.xlsx.

Protein AUC per classification task
Mann-Whitney AUC, Wilcoxon p (signed-rank within arm); 1900 features



Each feature contributes one AUC per task. Within-arm tasks compare a subject's later timepoint with its earlier one; HRvLR tasks compare subjects on the baseline level or on their own change. Dashed line: chance. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheets classify and chance_classify.

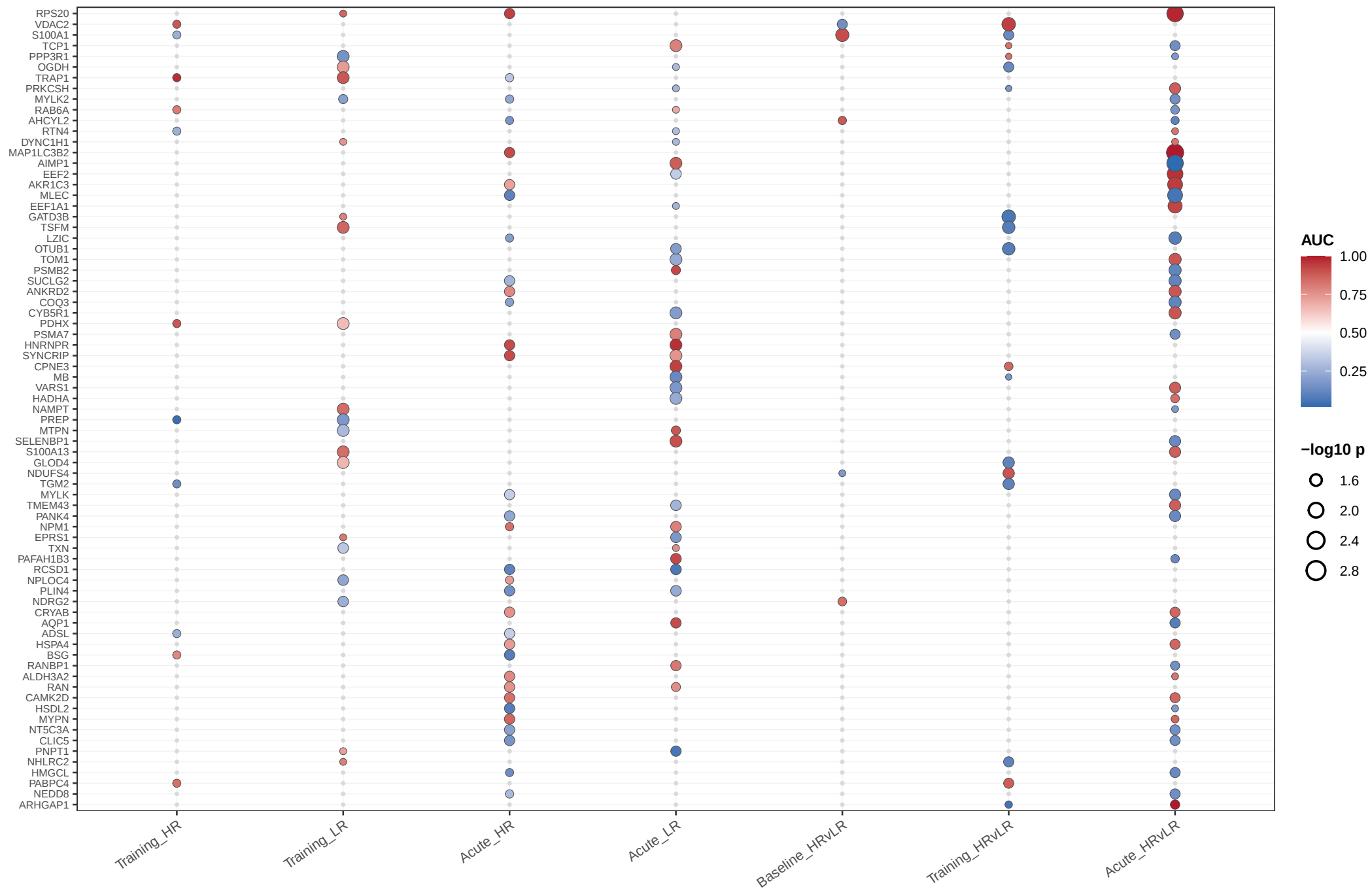
Protein association with phenotype
limma on a continuous predictor; BH per cell; 1574 features



Rows: the feature at T1, or its change over training (T2 – T1) or the acute bout (T3 – T2). Columns: the ten phenotypes. Fill: nominal hits at p < 0.05 over the count expected by chance (0.05 x tested); white is what an empty screen looks like. Top number: nominal hits; bottom: BH < 0.05. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheets associate and chance_associate.

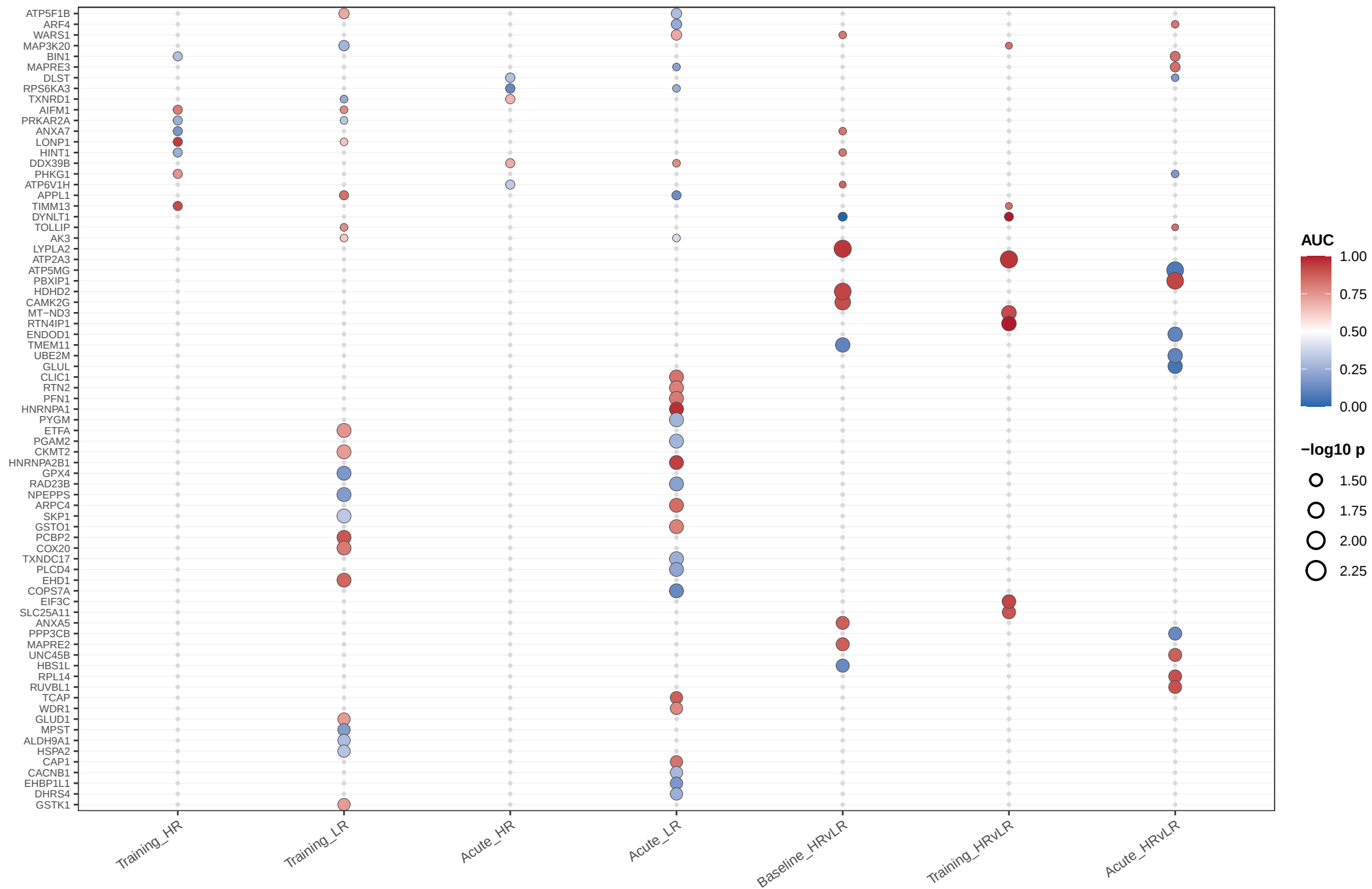
Protein classification hits (1/5)

AUC and Wilcoxon p per task, paired within arm; 351 features nominal in at least one column; rows 1–75



Protein classification hits (2/5)

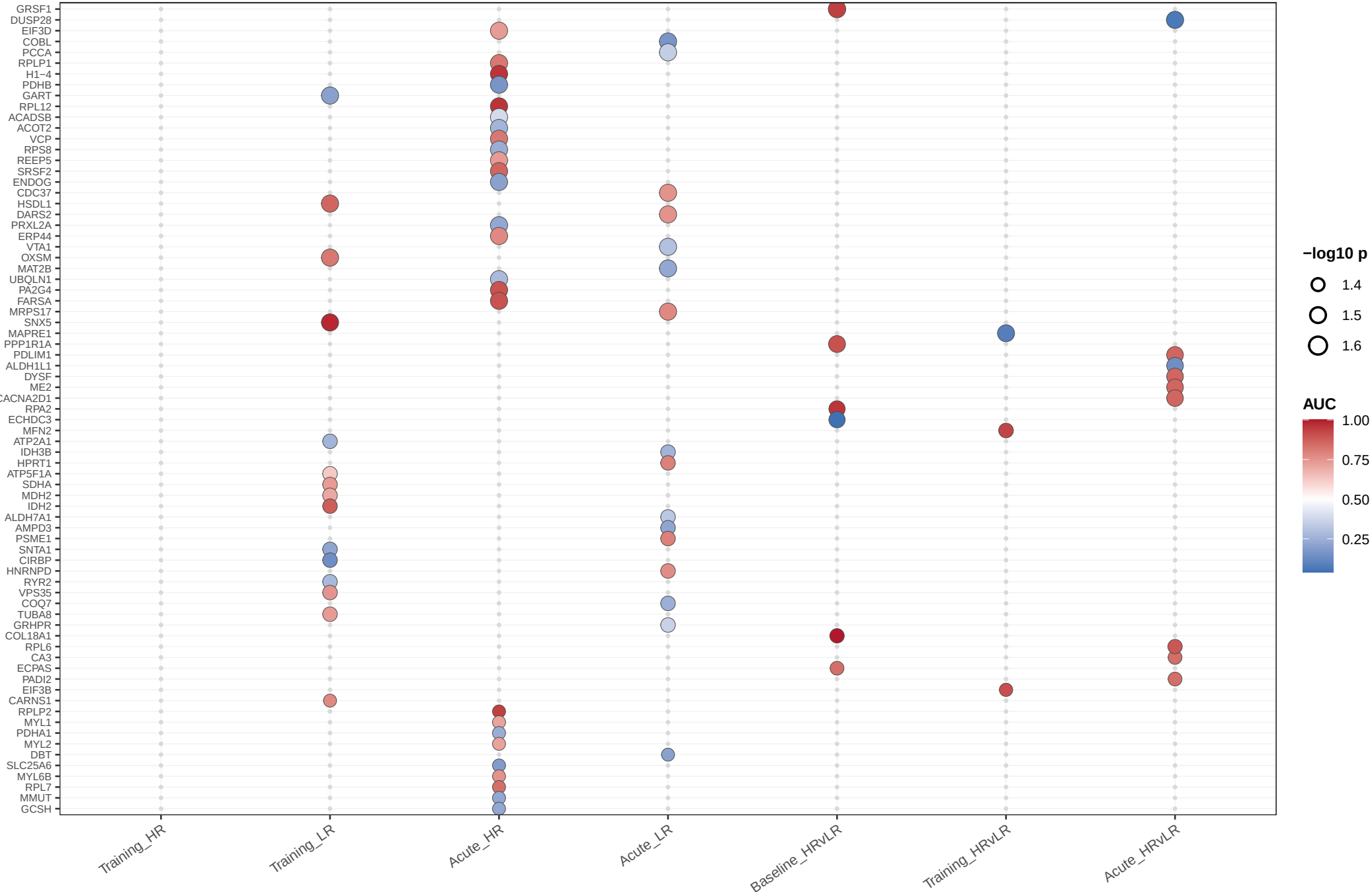
AUC and Wilcoxon p per task, paired within arm; 351 features nominal in at least one column; rows 76–150



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = AUC, size = $-\log_{10} p$. Black ring: BH < 0.05 . Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet classify.

Protein classification hits (3/5)

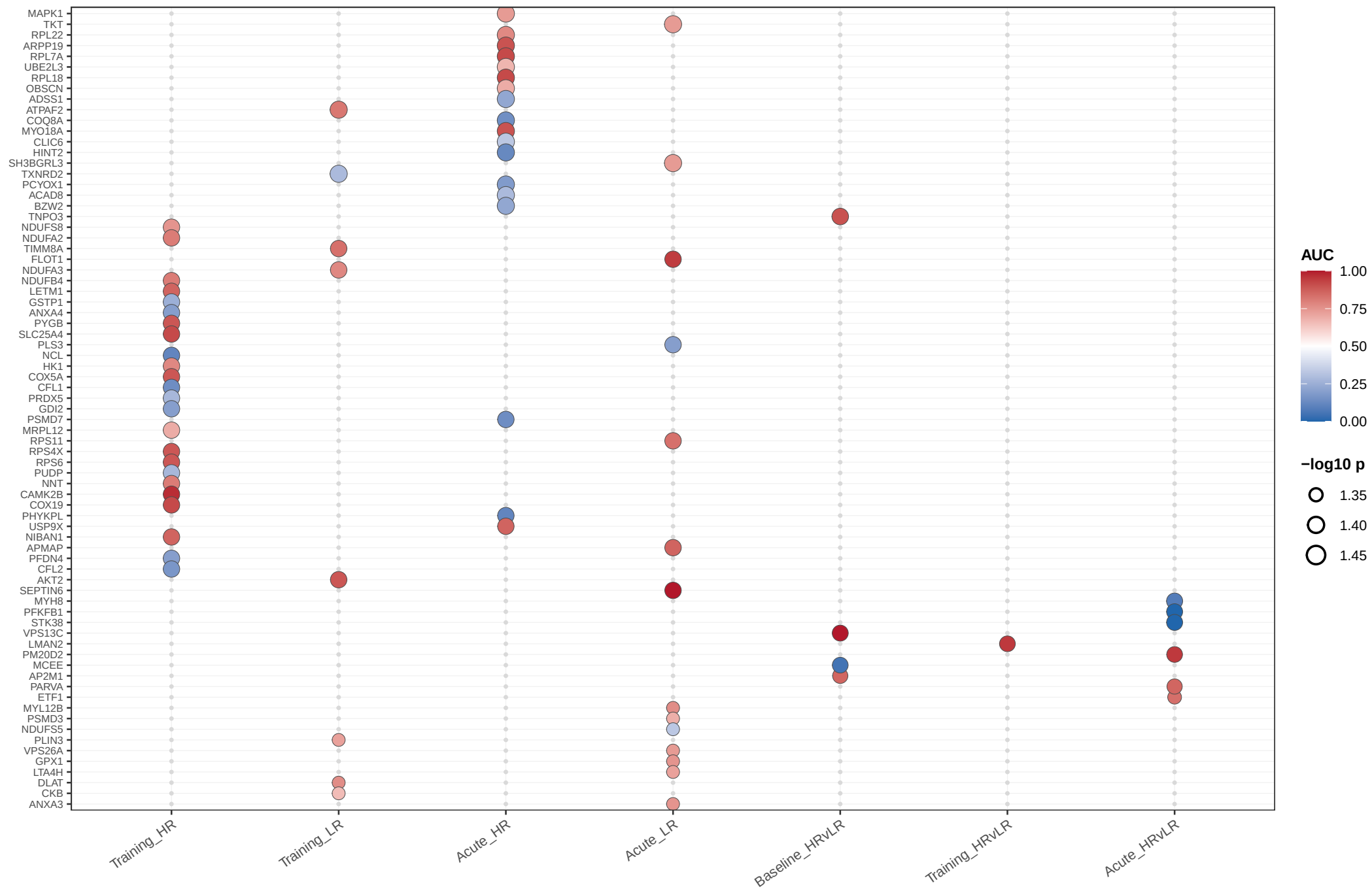
AUC and Wilcoxon p per task, paired within arm; 351 features nominal in at least one column; rows 151–225



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = AUC, size = -log₁₀ p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet classify.

Protein classification hits (4/5)

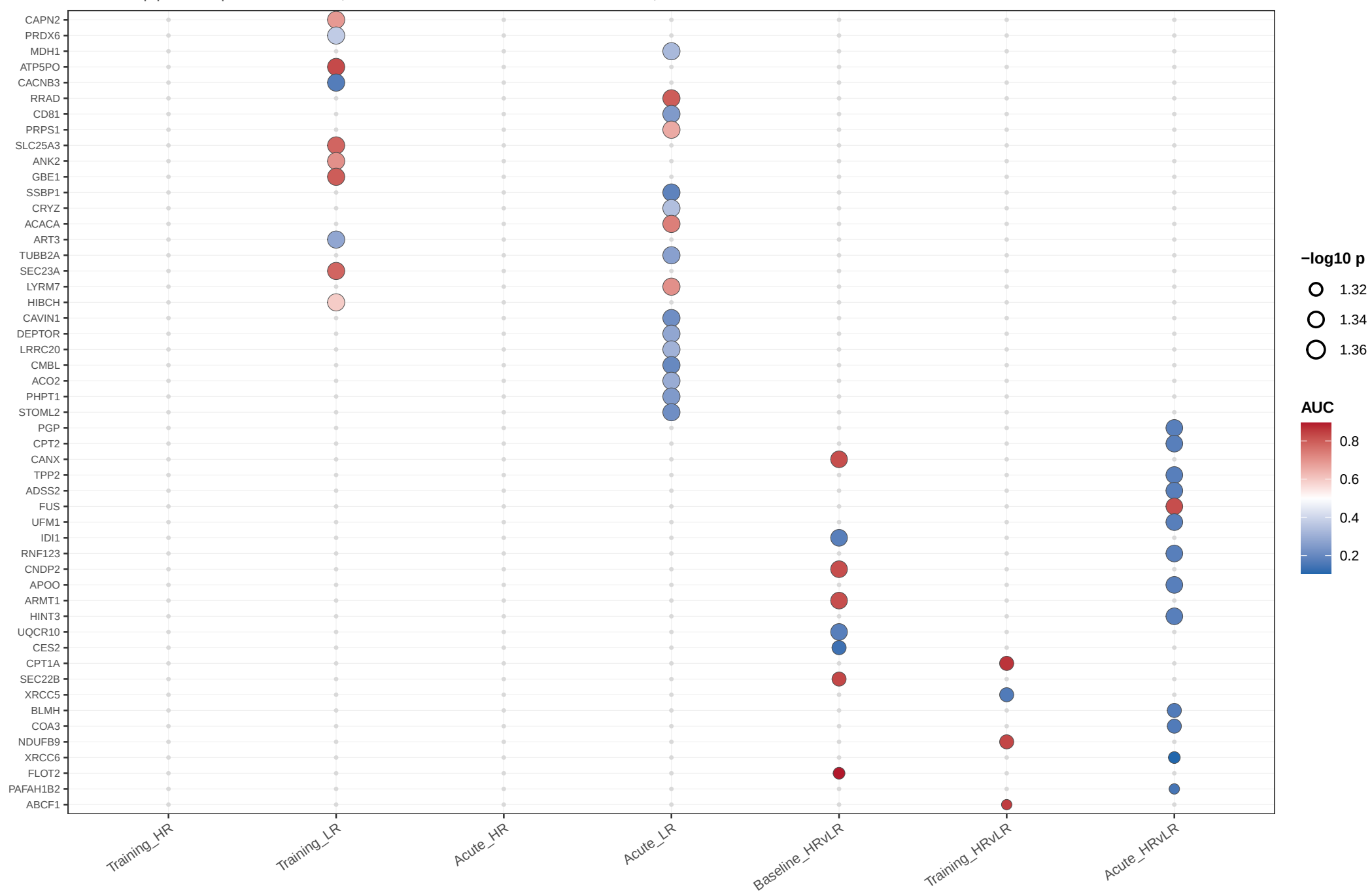
AUC and Wilcoxon p per task, paired within arm; 351 features nominal in at least one column; rows 226–300



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = AUC, size = $-\log_{10} p$. Black ring: BH < 0.05 . Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet classify.

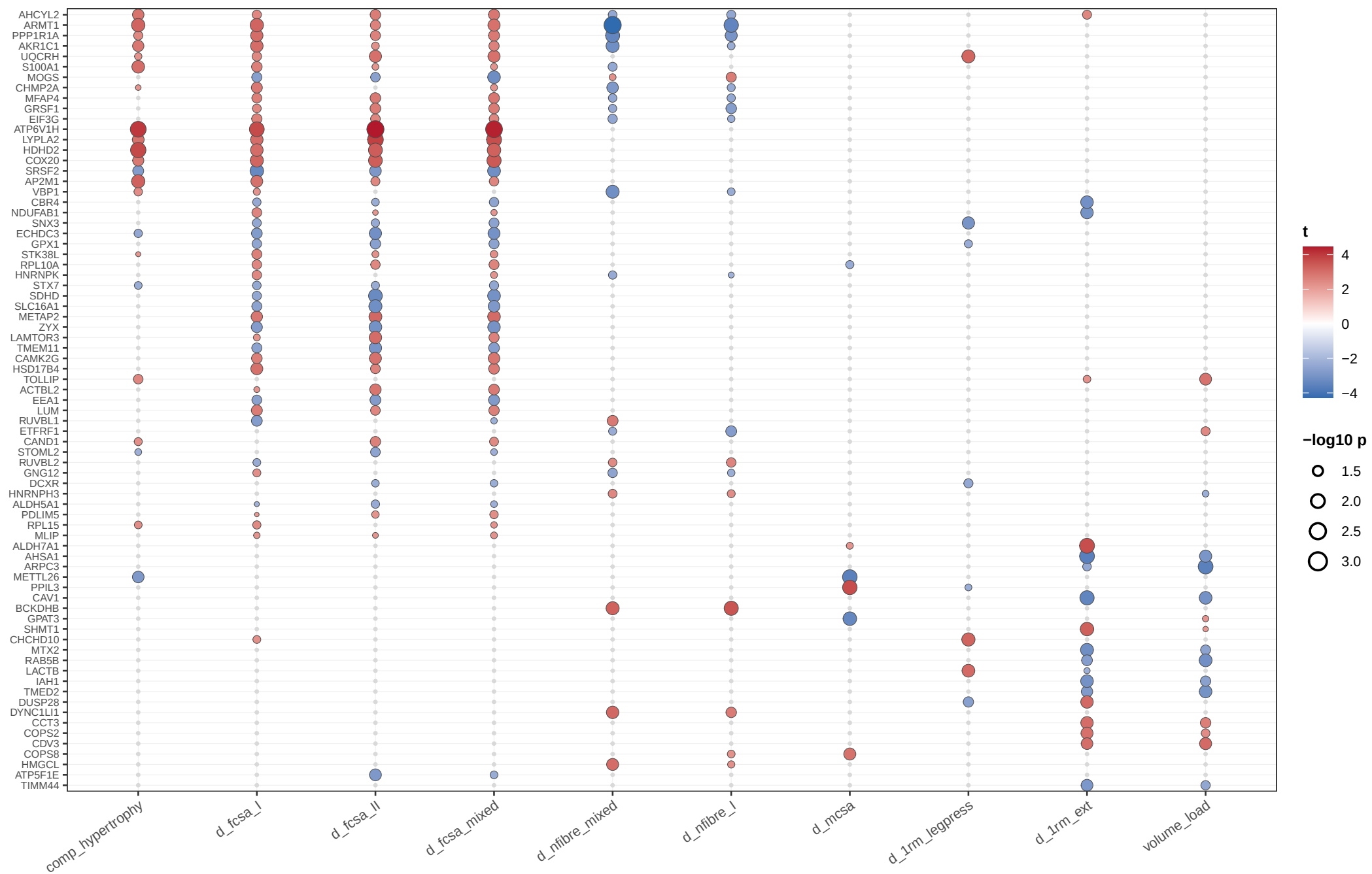
Protein classification hits (5/5)

AUC and Wilcoxon p per task, paired within arm; 351 features nominal in at least one column; rows 301–351



Protein association hits, level at T1 (1/7)

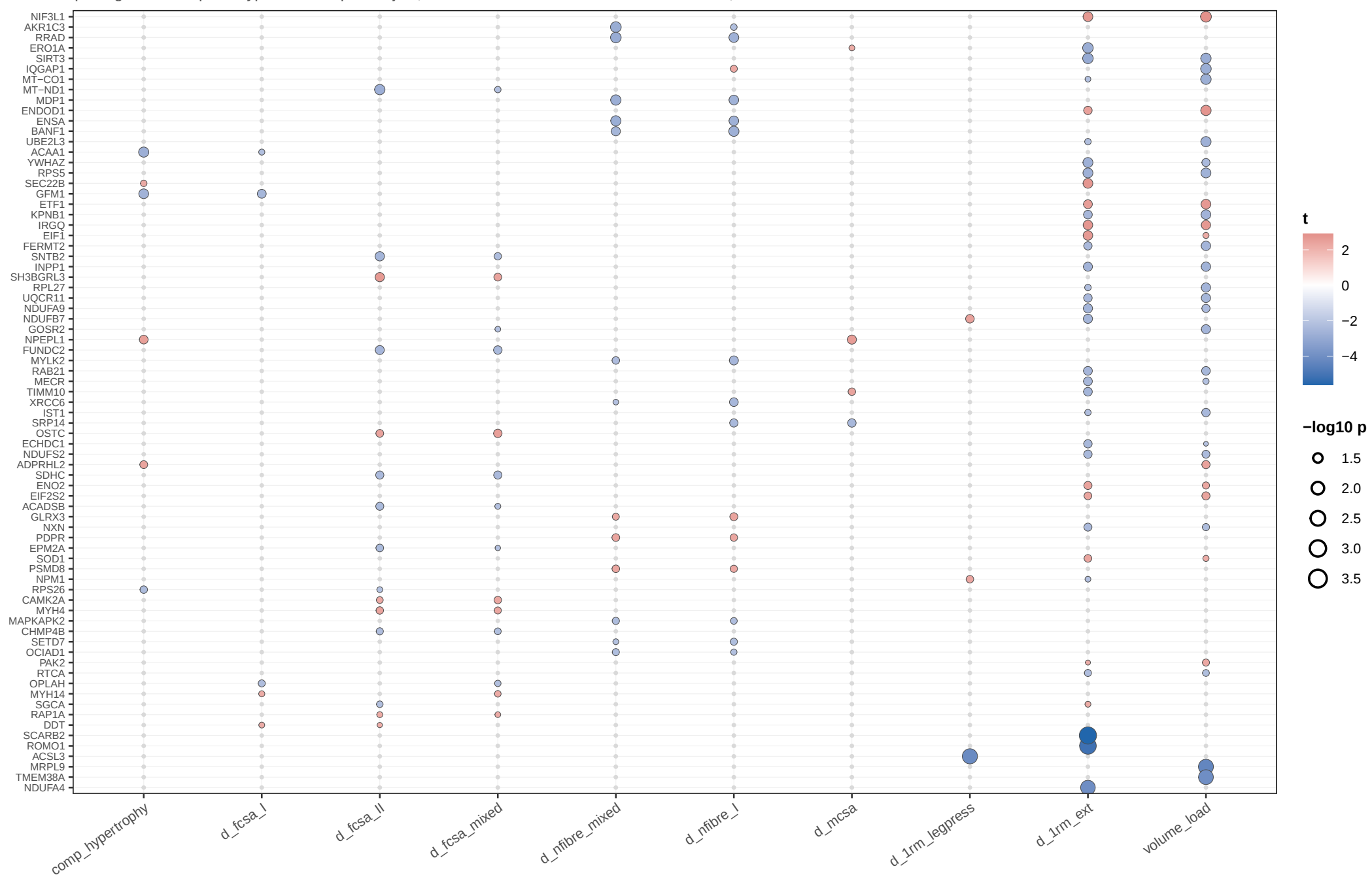
limma slope t against each phenotype, one row per subject; 452 features nominal in at least one column; rows 1–75



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05 . Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

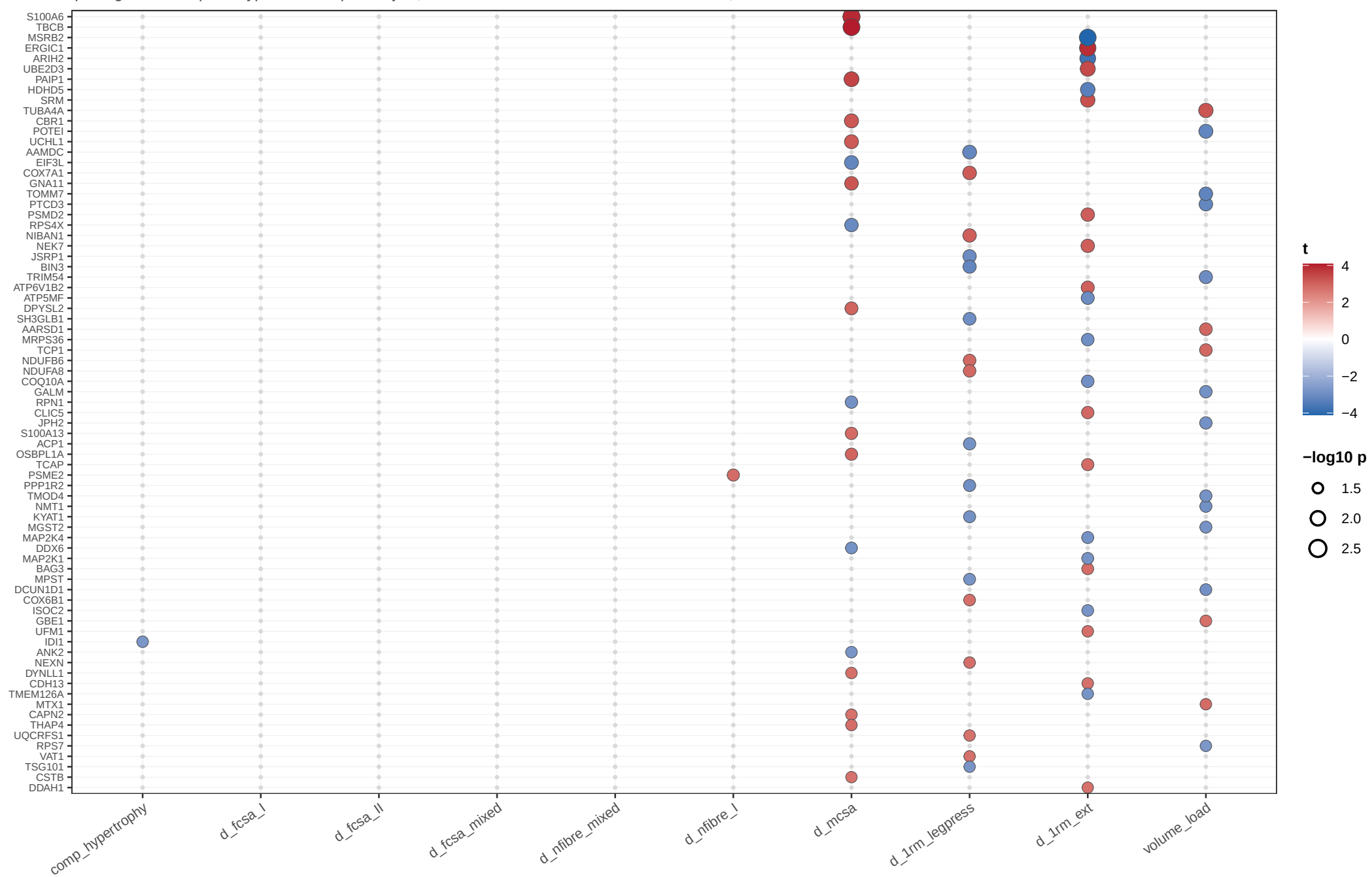
Protein association hits, level at T1 (2/7)

limma slope t against each phenotype, one row per subject; 452 features nominal in at least one column; rows 76–150



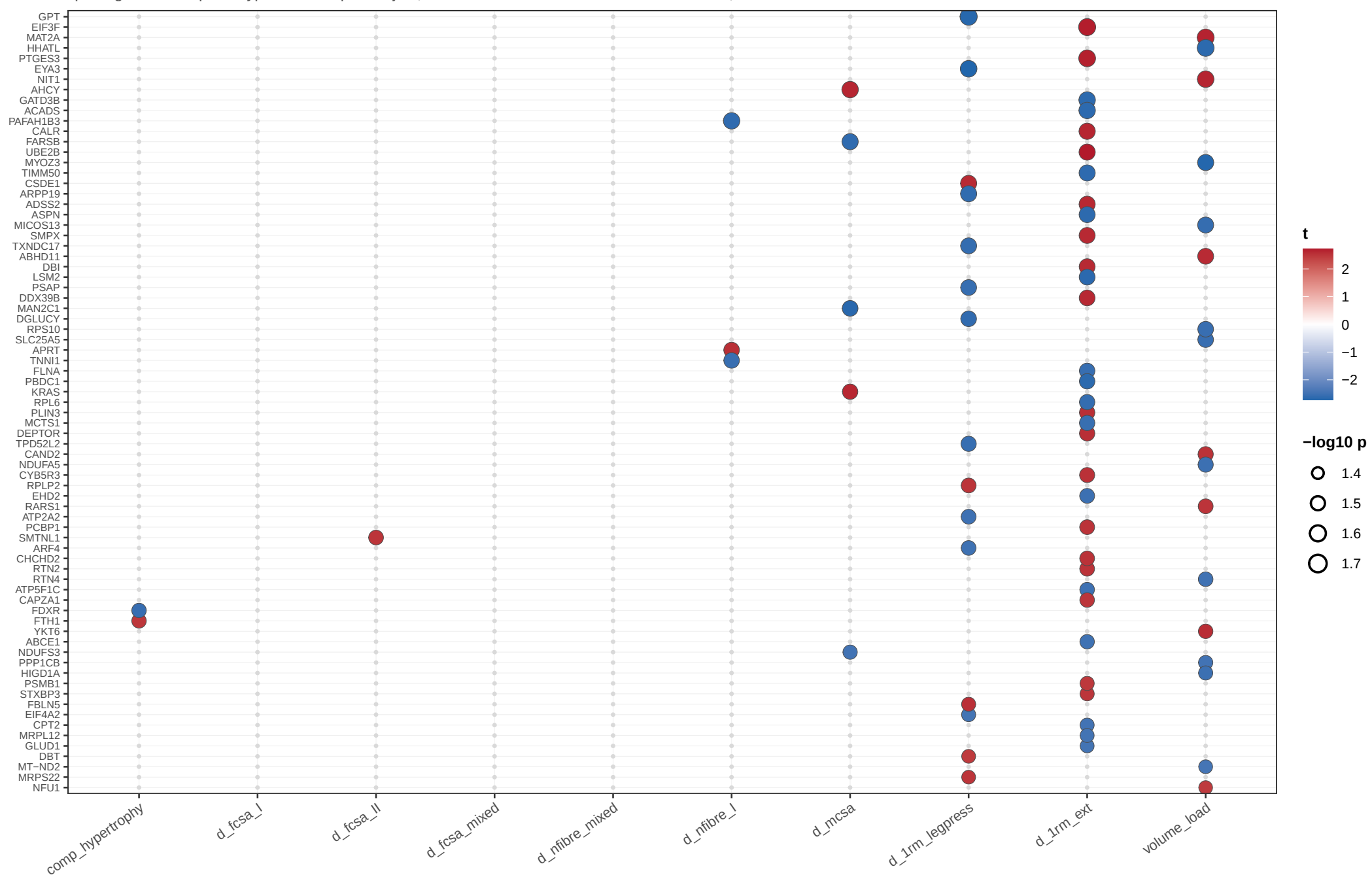
Protein association hits, level at T1 (3/7)

limma slope t against each phenotype, one row per subject; 452 features nominal in at least one column; rows 151–225



Protein association hits, level at T1 (4/7)

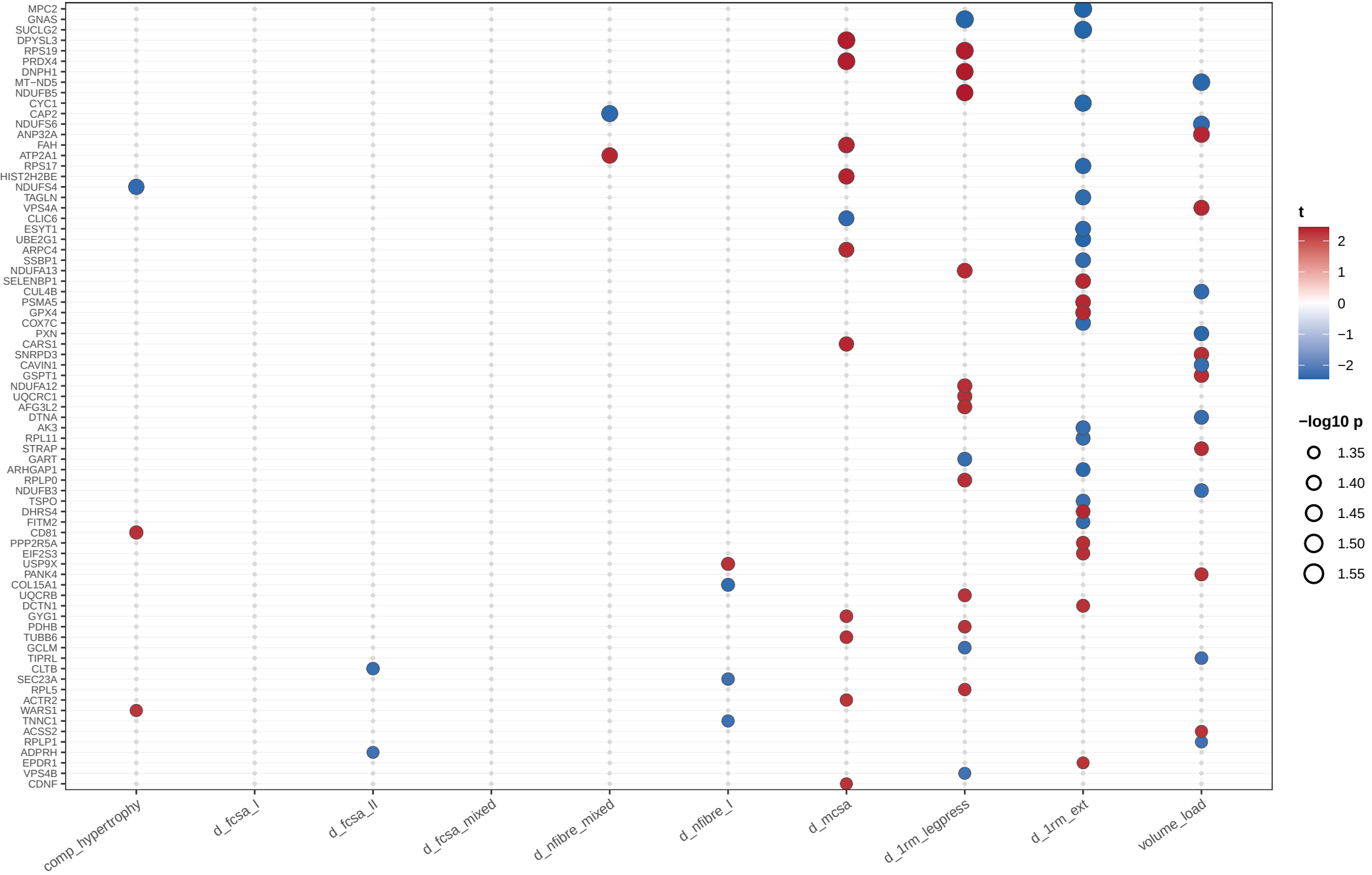
limma slope t against each phenotype, one row per subject; 452 features nominal in at least one column; rows 226–300



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, level at T1 (5/7)

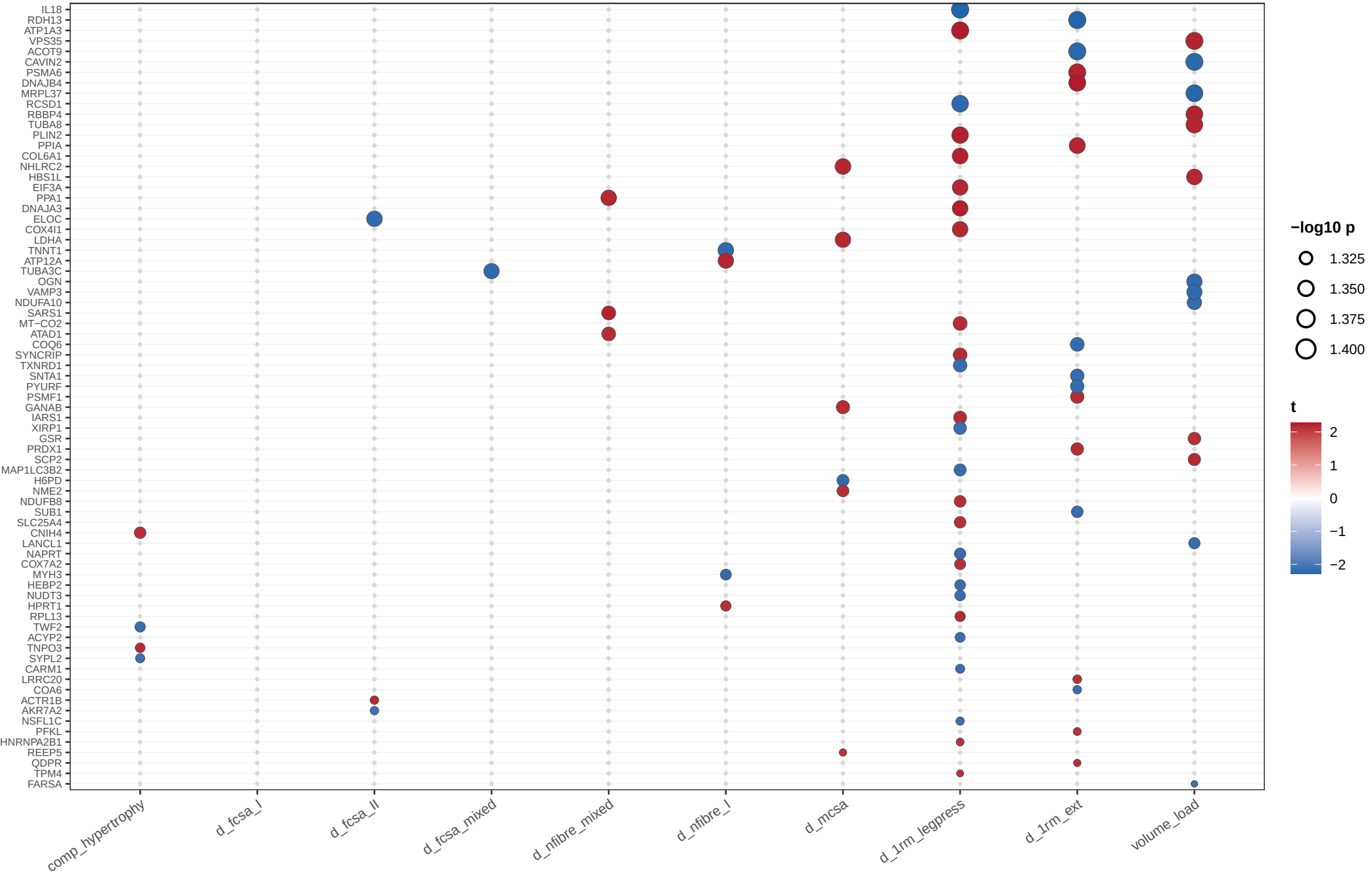
limma slope t against each phenotype, one row per subject; 452 features nominal in at least one column; rows 301–375



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = -log₁₀ p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

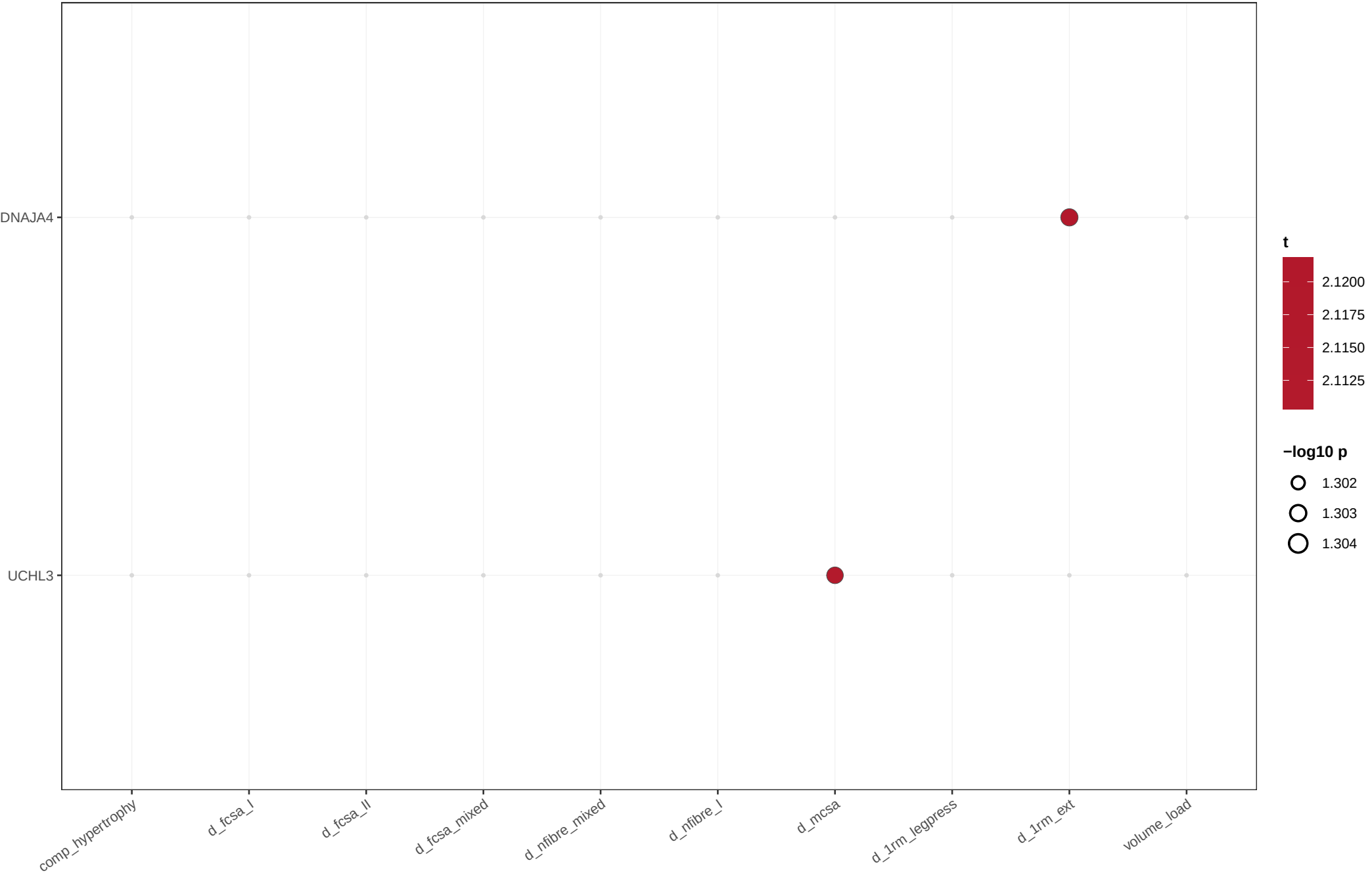
Protein association hits, level at T1 (6/7)

limma slope t against each phenotype, one row per subject; 452 features nominal in at least one column; rows 376–450



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = -log₁₀ p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

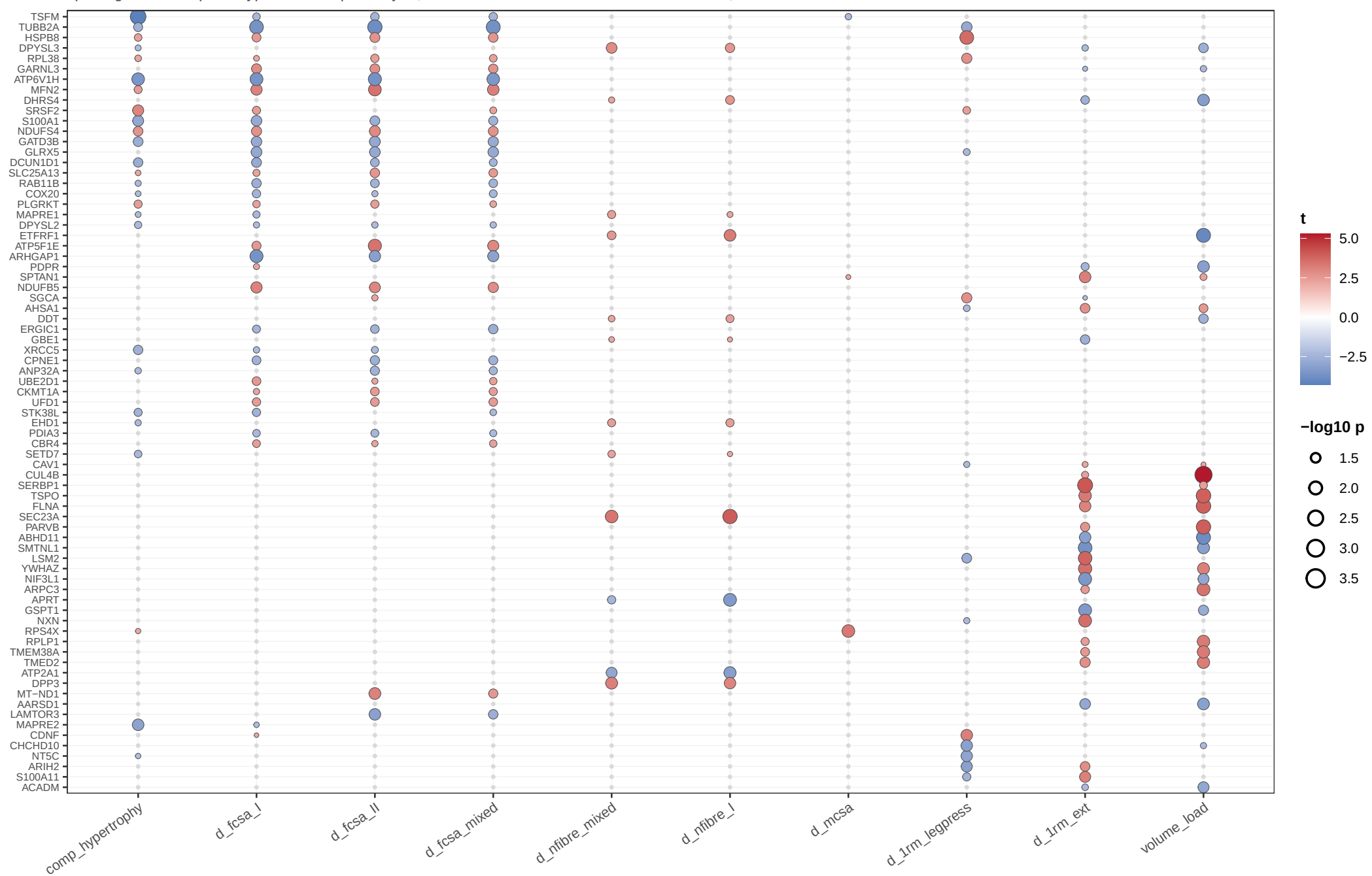
Protein association hits, level at T1 (7/7)
limma slope t against each phenotype, one row per subject; 452 features nominal in at least one column; rows 451–452



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05 . Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, training change (T2 – T1) (1/6)

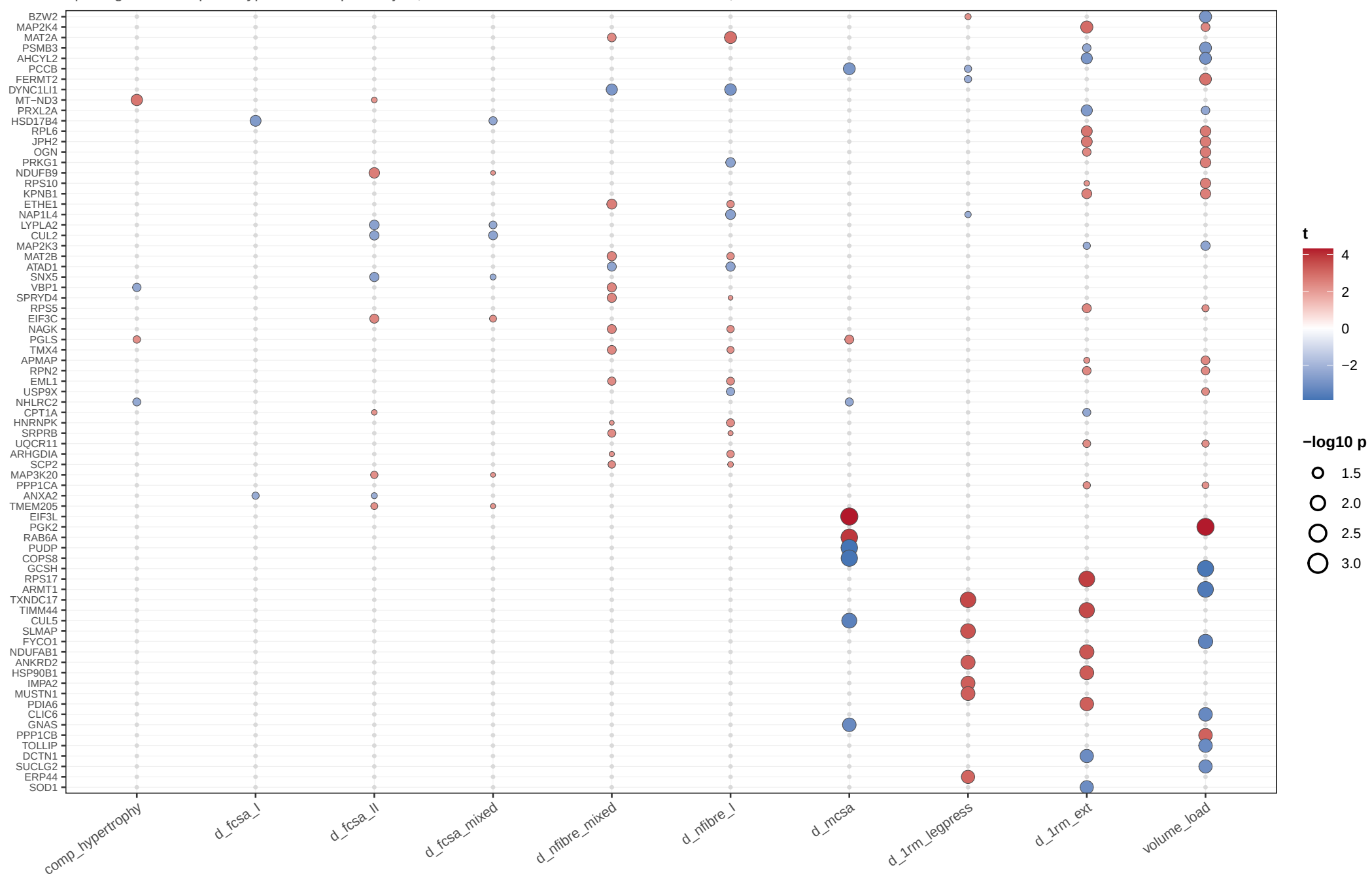
limma slope t against each phenotype, one row per subject; 385 features nominal in at least one column; rows 1–75



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05 . Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, training change (T2 - T1) (2/6)

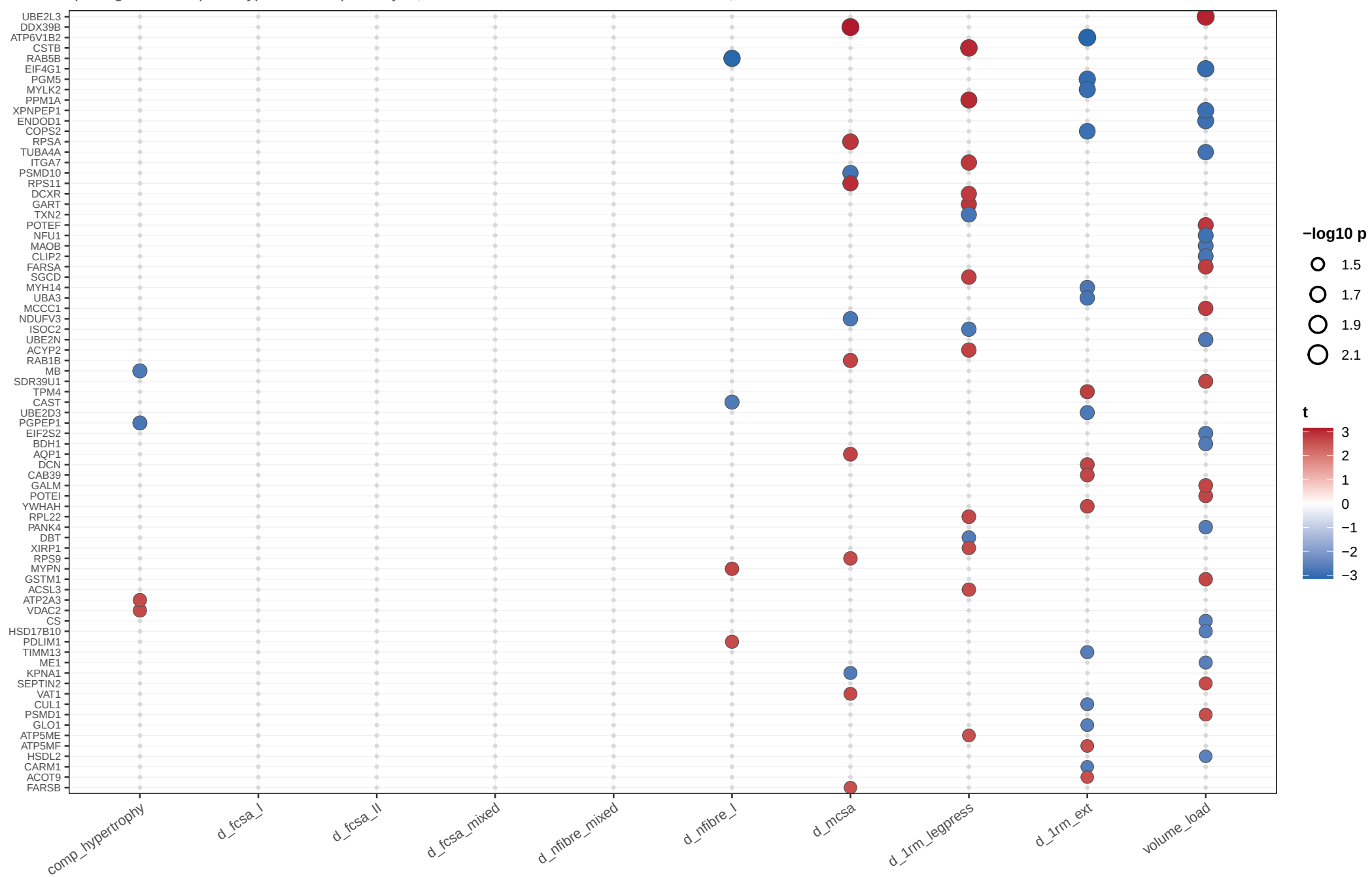
limma slope t against each phenotype, one row per subject; 385 features nominal in at least one column; rows 76-150



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05 . Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, training change (T2 - T1) (3/6)

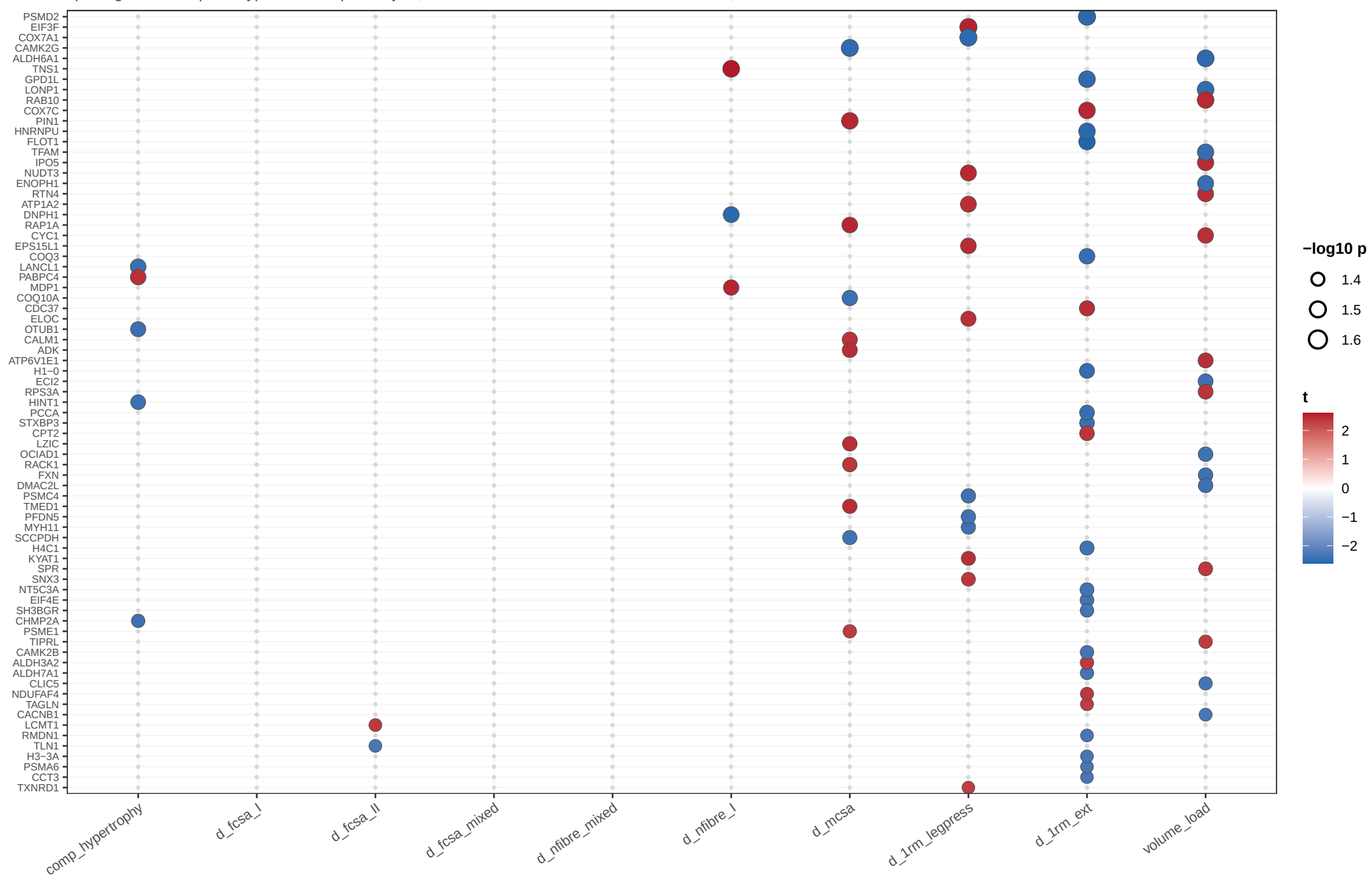
limma slope t against each phenotype, one row per subject; 385 features nominal in at least one column; rows 151–225



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, training change (T2 – T1) (4/6)

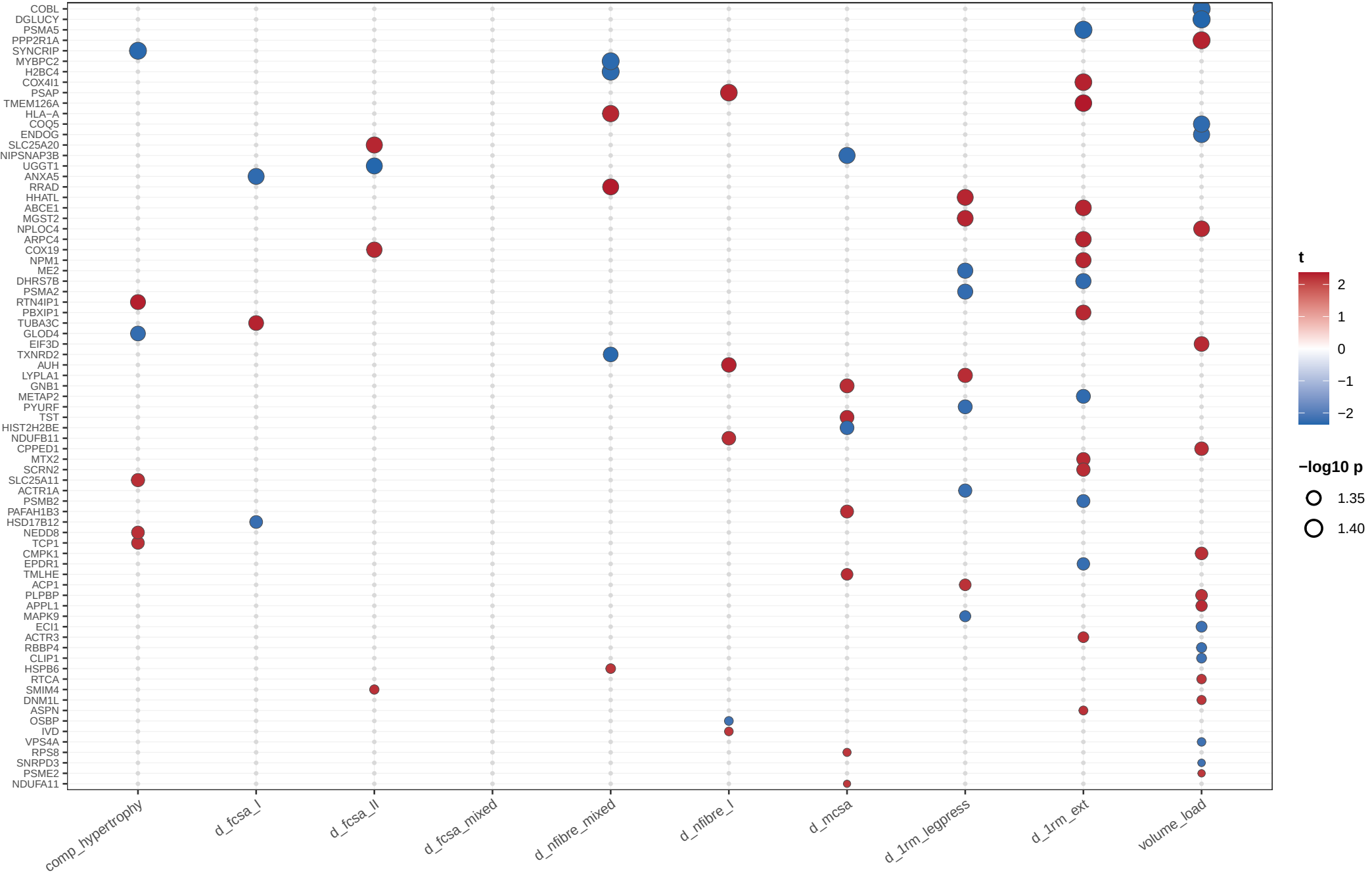
limma slope t against each phenotype, one row per subject; 385 features nominal in at least one column; rows 226–300



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, training change (T2 - T1) (5/6)

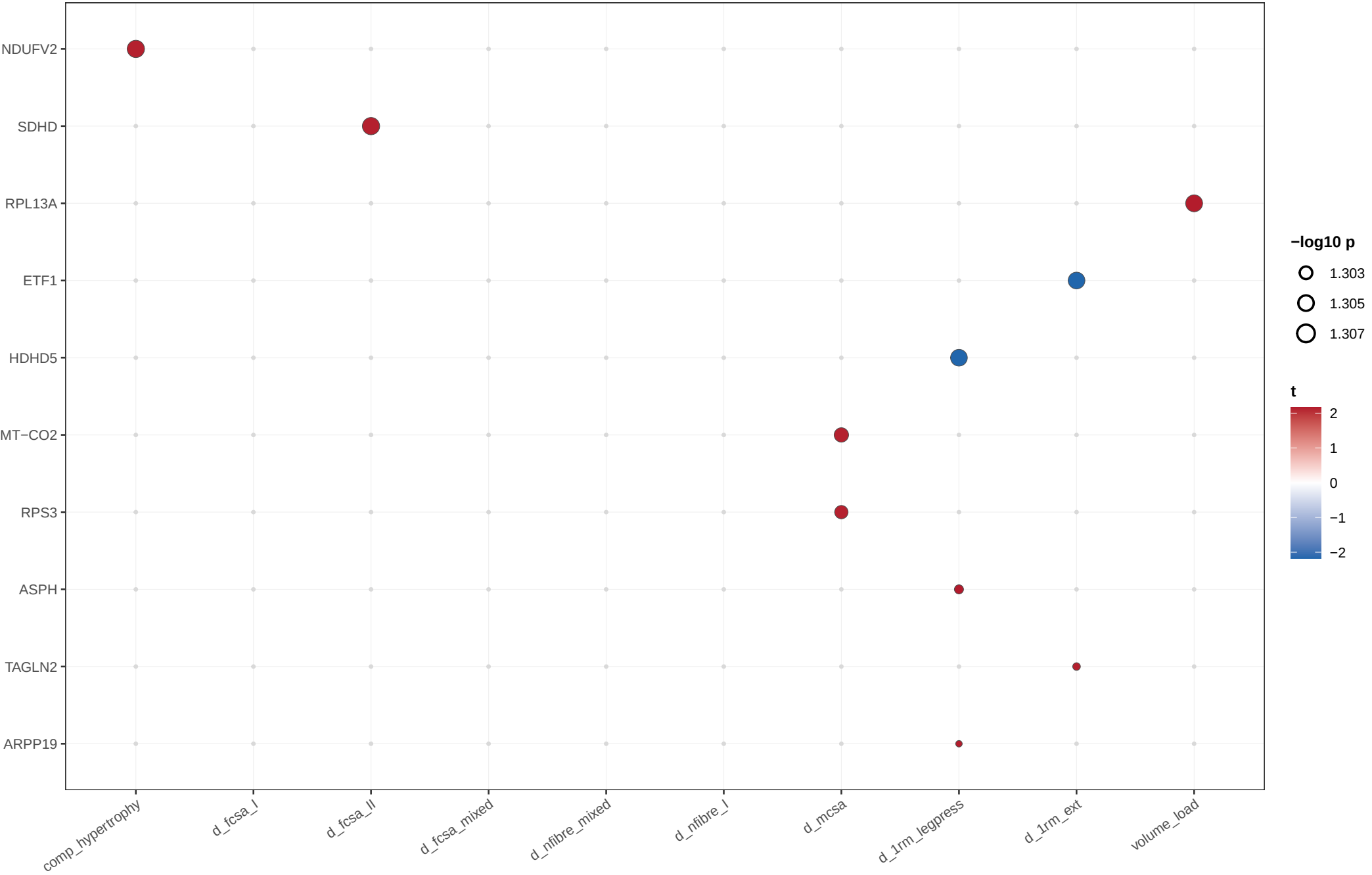
limma slope t against each phenotype, one row per subject; 385 features nominal in at least one column; rows 301-375



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = $-\log_{10} p$. Black ring: $BH < 0.05$. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, training change (T2 – T1) (6/6)

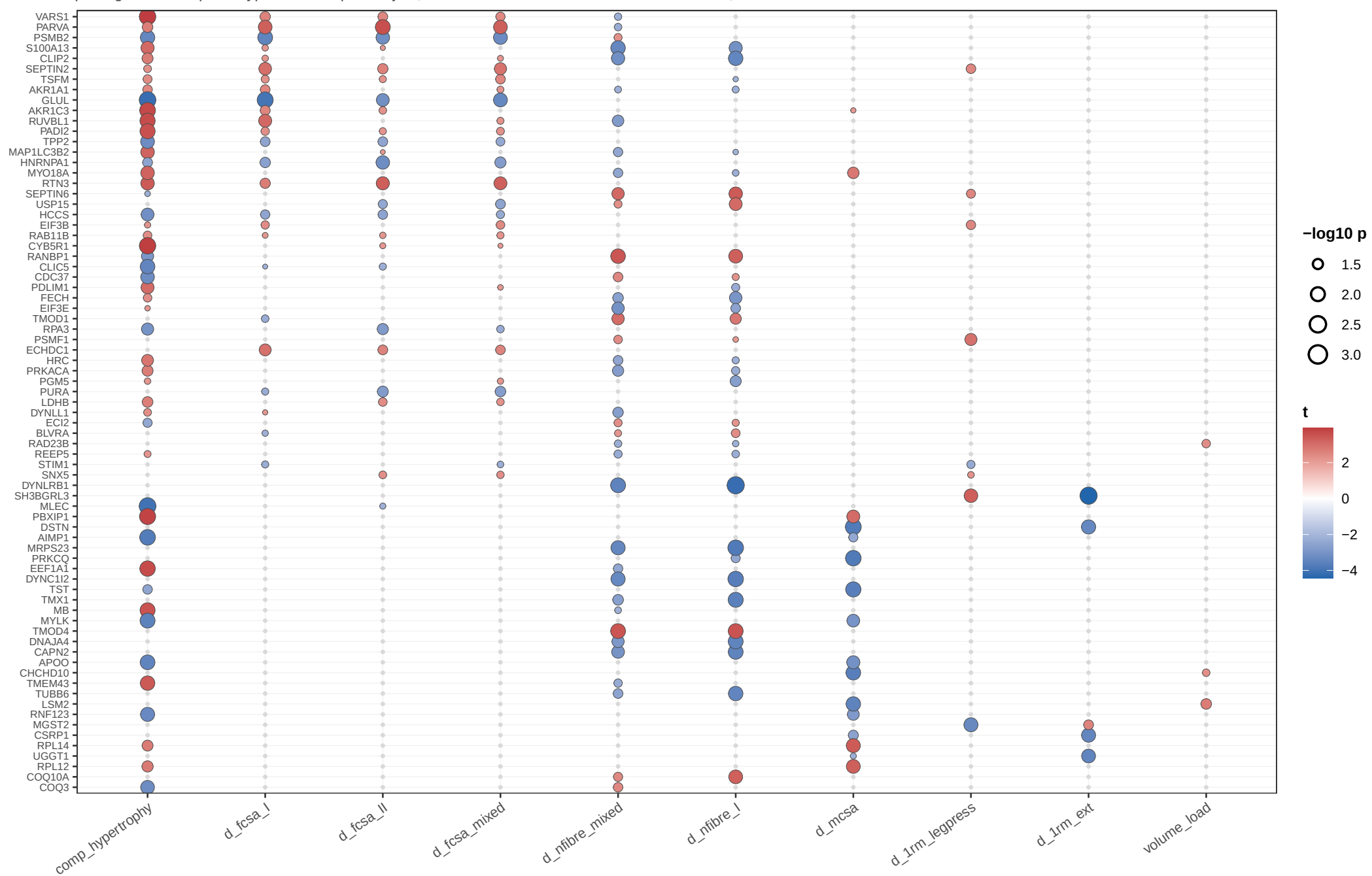
limma slope t against each phenotype, one row per subject; 385 features nominal in at least one column; rows 376–385



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = -log10 p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, acute change (T3 – T2) (1/7)

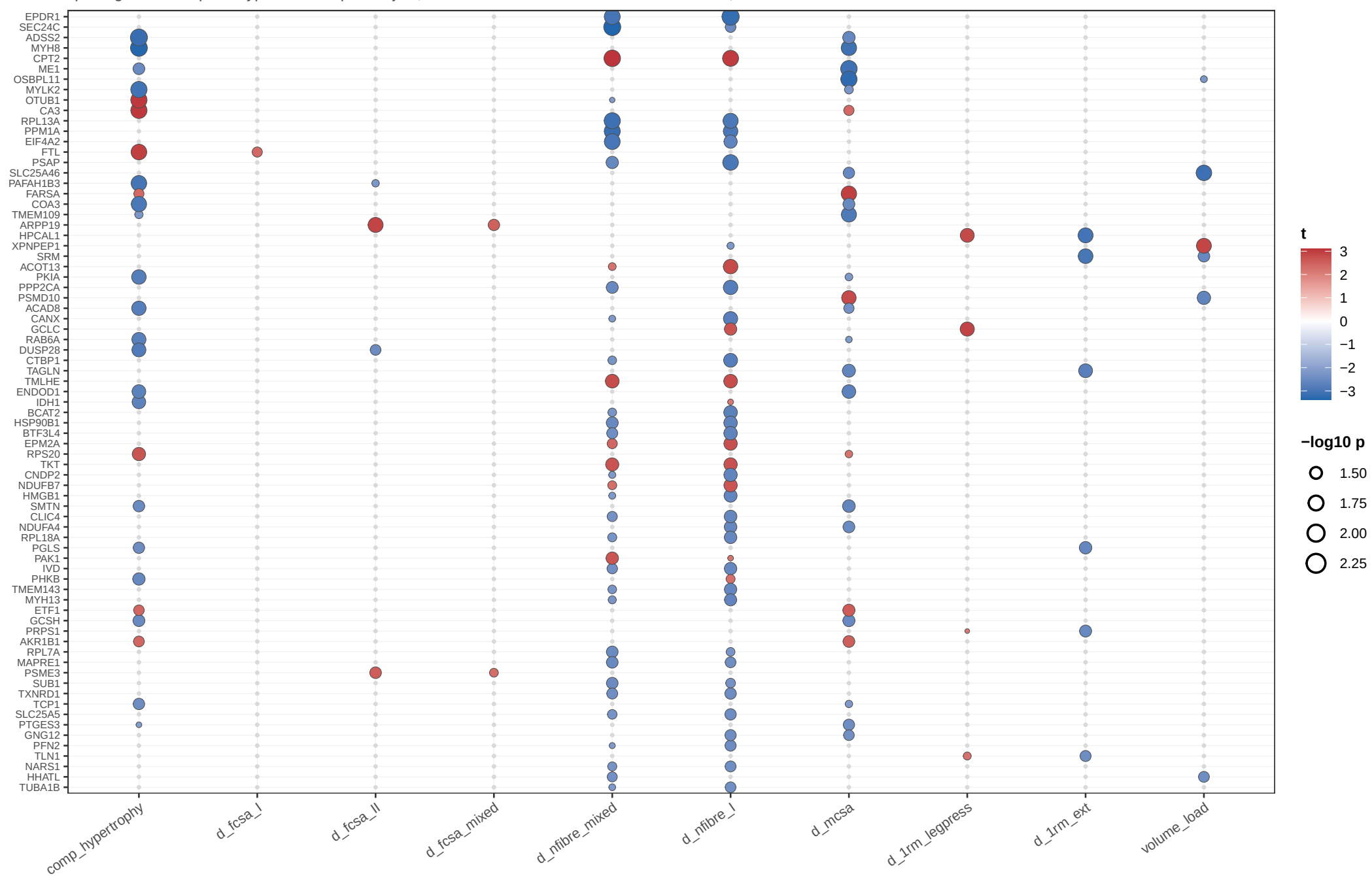
limma slope t against each phenotype, one row per subject; 482 features nominal in at least one column; rows 1–75



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, acute change (T3 – T2) (2/7)

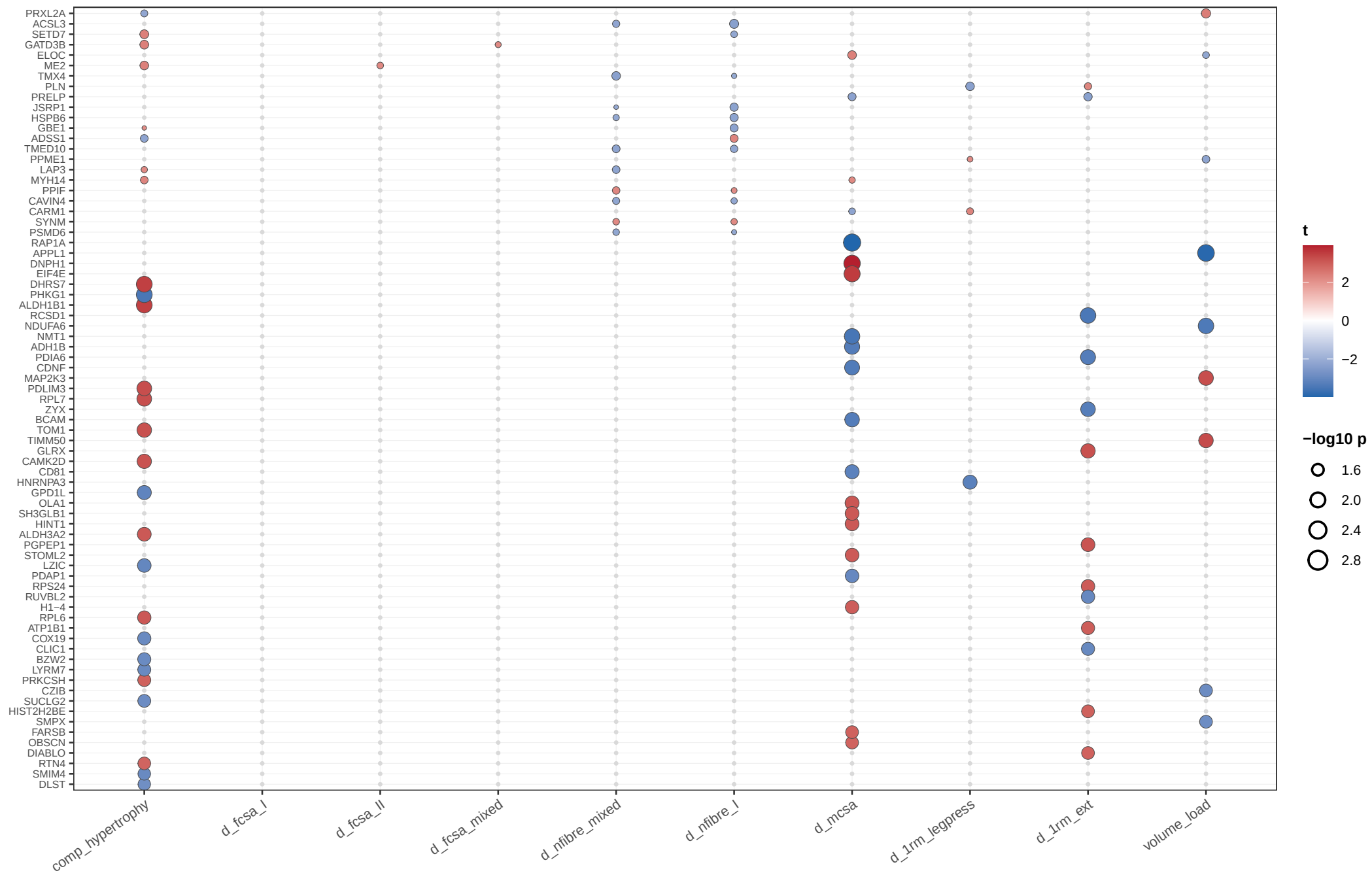
limma slope t against each phenotype, one row per subject; 482 features nominal in at least one column; rows 76–150



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05 . Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, acute change (T3 – T2) (3/7)

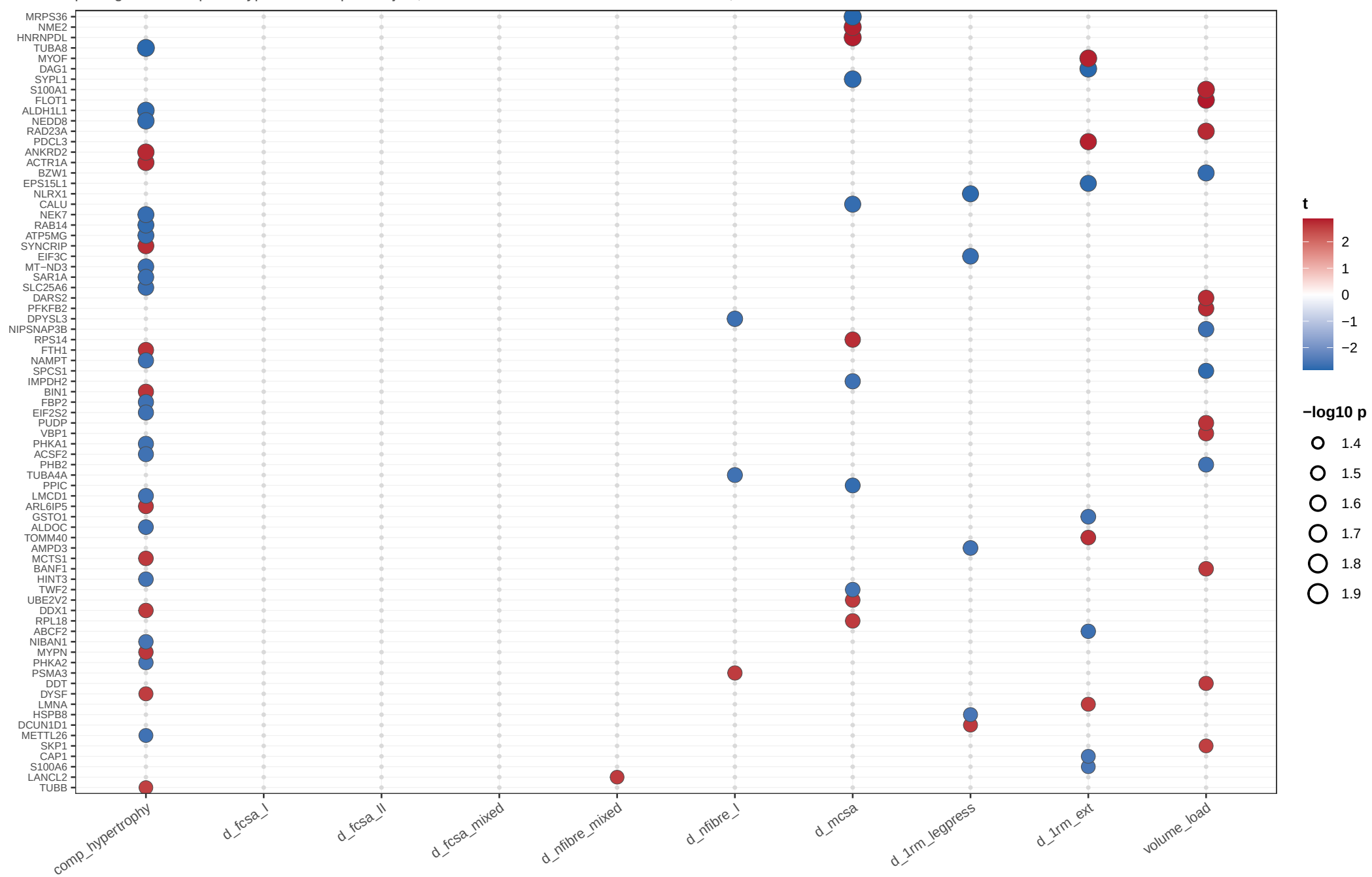
limma slope t against each phenotype, one row per subject; 482 features nominal in at least one column; rows 151–225



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, acute change (T3 – T2) (4/7)

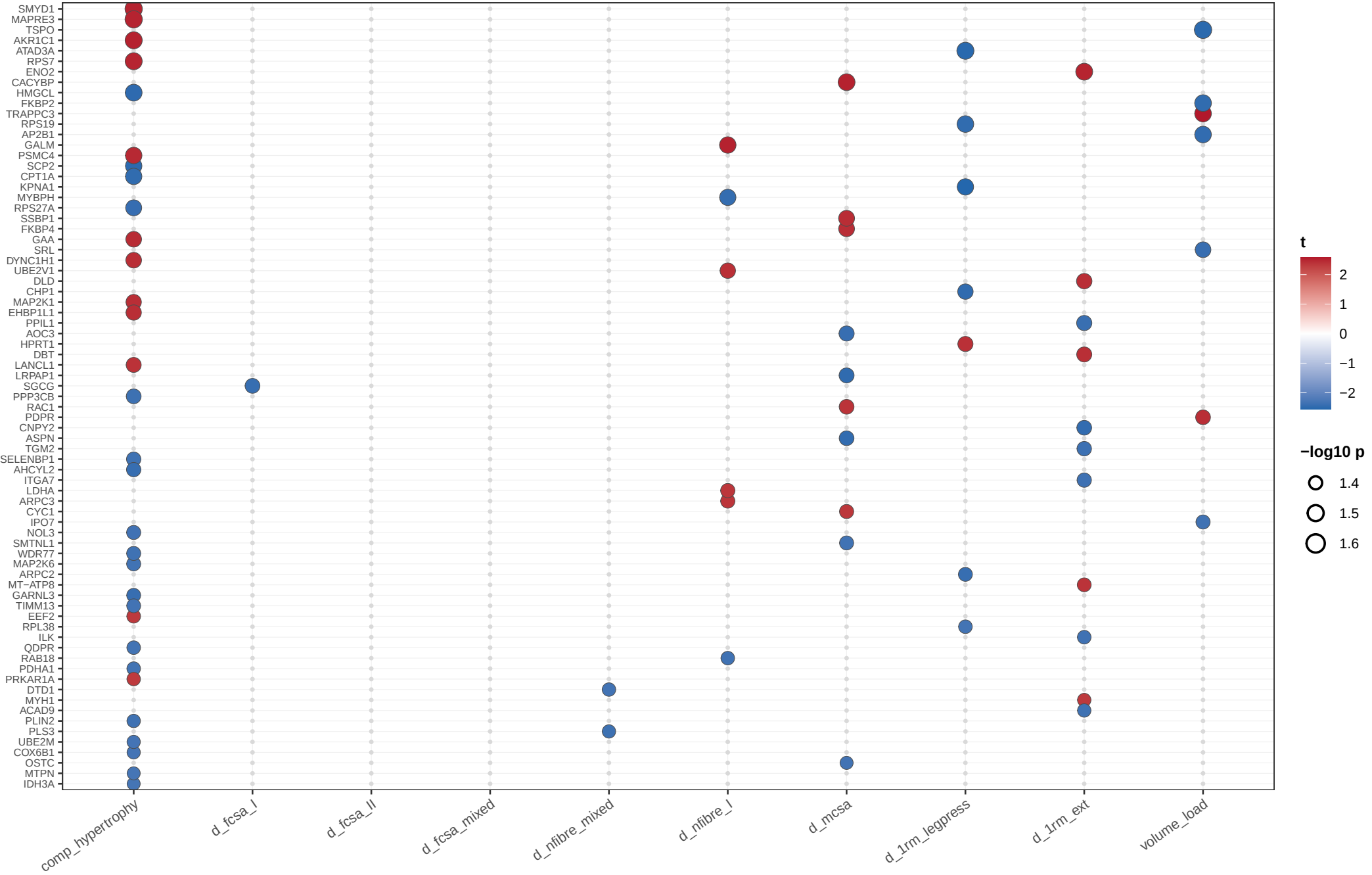
limma slope t against each phenotype, one row per subject; 482 features nominal in at least one column; rows 226–300



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05 . Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, acute change (T3 – T2) (5/7)

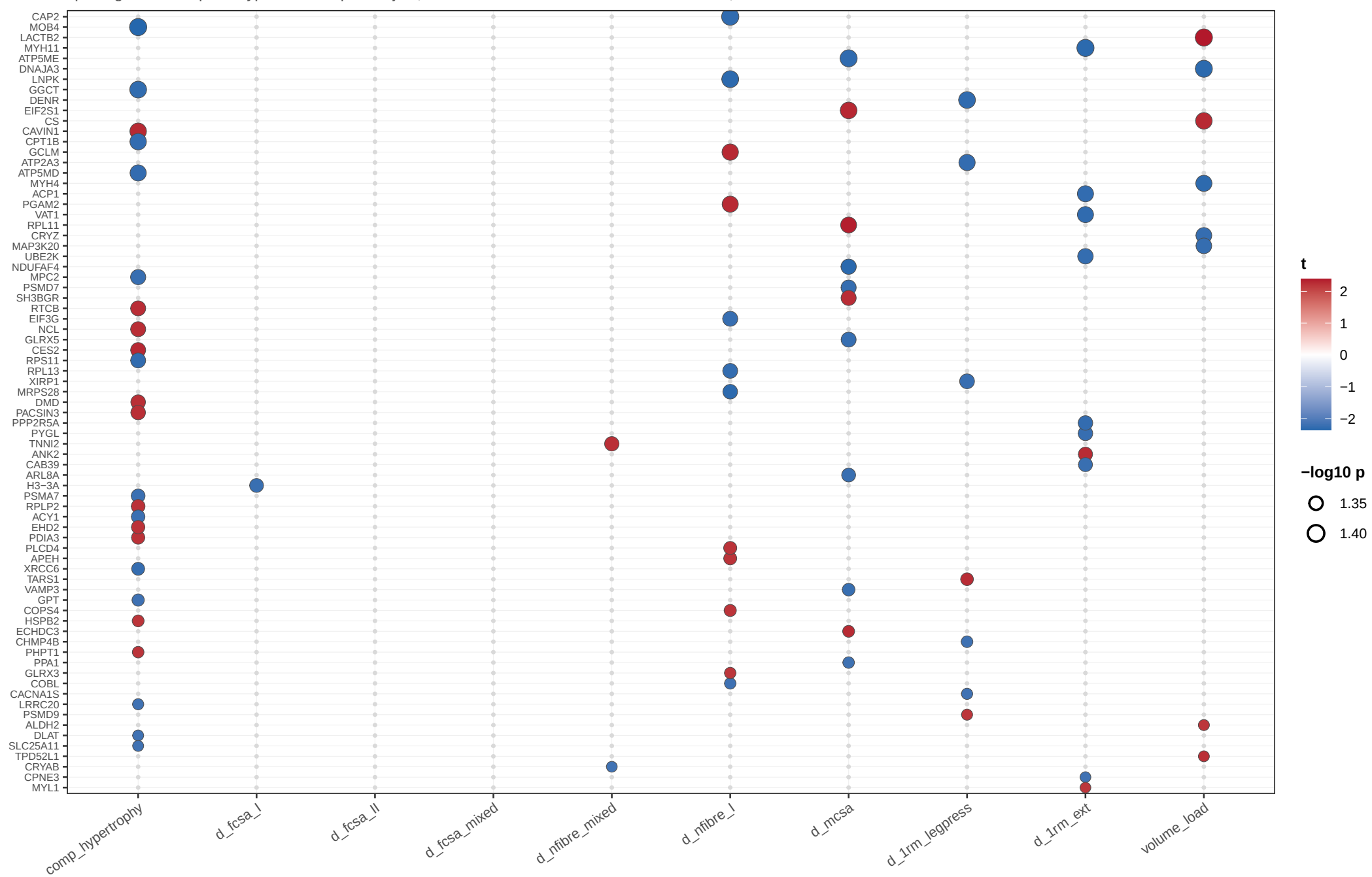
limma slope t against each phenotype, one row per subject; 482 features nominal in at least one column; rows 301–375



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = -log₁₀ p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, acute change (T3 – T2) (6/7)

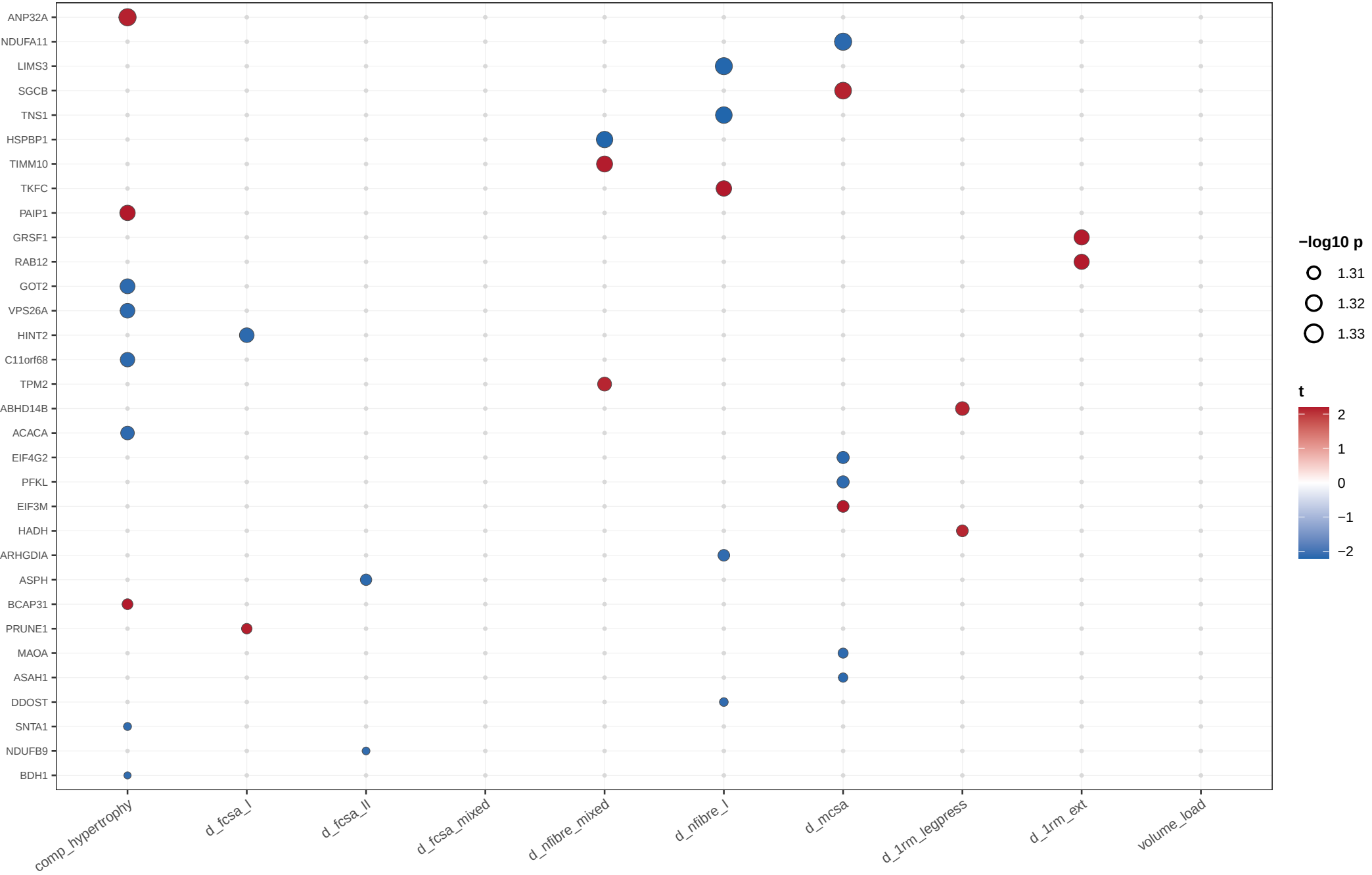
limma slope t against each phenotype, one row per subject; 482 features nominal in at least one column; rows 376–450



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.

Protein association hits, acute change (T3 – T2) (7/7)

limma slope t against each phenotype, one row per subject; 482 features nominal in at least one column; rows 451–482



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = -log₁₀ p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 02_Proteins/02_Classify_Associate/c_data/protein_screens.xlsx, sheet associate.